Updated computations for the examples of Rook_Field_Paper_v2

Recomputed from the manuscript's own network specifications and parameter nodes

July 28, 2026

Summary

All 17 computational examples in the monograph were recomputed under two algorithm specifications:

- **legacy** — reproduces the current DSGRN_UTILS bit-for-bit (verified across five levels and four network families), so it matches whatever produced the published figures;
- **paper** — the live manuscript definitions.

30 of 34 runs match the printed values exactly. The remaining 4 are analysed below; each is understood, and none is a computational error in the manuscript's mathematics.

All four large networks reproduce their stated parameter-graph sizes exactly (87,280,405,632; 3,600,000; 13,608,000,000; 4,429,771,960,320), and both directly specified ramp systems reproduce their stated index sets. Runtimes are comparable to or faster than the reported ones.

At a glance

Example	Level	legacy	paper	Note
ex:saddle_saddle_3D_intro	3	ok	ok	
ex:introPeriodic	3	ok	ok	
ex:saddlesaddlebif	3	ok	ok	
ex:saddle_saddle_3D_example	3	ok	ok	
ex:periodicOrbit3	3	ok	ok	
ex:periodicOrbit3_F3	3	ok	differs	**paper** resolves the component (M4)
ex:mccord	3	ok	ok	
ex:3d_example_1	3	ok	ok	
EMT-6D-semiconjugacy	2	ok	ok	
ex:trivial_index_2D	3	ok	ok	
ex:trivial_index_3D_F3	3	differs	ok	published 61/12 reproduces only under **paper** (M1)
ex:trivial_indices	3	ok	ok	
EMT-6D-trivial-indices	2	ok	ok	count matches; node labels are not canonical (M8)
ex:ramp_van_der_pol	3	ok	ok	cycle rule never fires: $\mathcal{F}_3 = \mathcal{F}_2$ (M5)
ex:periodicOrbit3_F4	4	differs	differs	index set misprinted (M2)
ex:periodicOrbit5	4	ok	ok	

Example	Level	legacy	paper	Note
ex:missing_periodic_3D_F4	4	ok	ok	

Problems to address

Ordered by consequence. Items **M***n* are manuscript-side, items **C***n* are DSGRN_UTILS-side. Several manuscript symptoms have a code-side cause, and are cross-referenced.

Manuscript

M1 — ex:trivial_index_3D_F3: 61/12 no longer reproduces

examples4.tex:786 *(breaks a published number)*
The text reports a trivial-index component with 61 cells and 12 double-edge pairs. Recomputing with the current DSGRN_UTILS gives **75 cells and 16 double-edge pairs**. Under the live `def:active_regulation` the published 61/12 reproduces exactly.
So the published figure was produced with the manuscript's definition, and the code has since drifted away from it. Nothing in the manuscript needs changing here — the *code* does (see C1). The number is correct as printed.

M2 — ex:periodicOrbit3_F4: the index set is wrong in its first factor

examples4.tex:920--923 *(misprint)*
The displayed index set is

$$\mathcal{I} = \{0,1,2,3,4\} \times \{0,1,2,3,4\} \times \{0,1,2,3\}.$$

Node 1 of that network has two out-edges, so $K(1) = 2$ and the first factor is $\{0,1,2,3\}$. Computed: $\mathcal{I} = \{0,\dots,3\} \times \{0,\dots,4\} \times \{0,\dots,3\}$.
Everything else in the example — the 12-cell cycle, the absence of double edges, the periodic-orbit index — reproduces exactly.
Fix: change the first factor to $\{0,1,2,3\}$.

M3 — The published ramp parameters violate $\mathcal{H}_1$, hence $\mathcal{H}_2$ and $\mathcal{H}_3$

examples4.tex:857 (van der Pol), introduction6.tex:525 (3D periodic) *(hypotheses fail at the stated values)*
Both directly specified ramp systems satisfy $\mathcal{H}_0$ and $\Lambda(R)$ but **fail** $\mathcal{H}_1$: the focal value $E_n(D_{\mathbf{v}})/\gamma_n$ lands inside a ramp half-window, which `defn:H1` forbids. For van der Pol at box $(1,1)$, $E_1(D_{\mathbf{v}}) = 6.1$ and $\gamma_1 = 2$, so the focal value is 3.05, inside the forbidden window $(\theta_{m_1}, \theta_{m_1} + h_{m_1}] = (3, 3.15]$. Eight such violations for van der Pol, eighteen for the three-dimensional system.
Since $\mathcal{H}_2, \mathcal{H}_3 \subset \mathcal{H}_1$, the hypotheses of `thm:R1ABlattice` and `thm:R3ABlattice` do not hold at the stated values. `ex:introPeriodic` invokes `thm:dynamics` at exactly those values.
Fix: a uniform half-width $h = 0.005$ satisfies $\mathcal{H}_0$, $\Lambda(R)$, $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_3$, and *leaves the Morse graph unchanged* — same Conley indices, same Morse-graph edges. Only the two parameter tables need new h values; every computed conclusion stands.

M4 — Under the live definition, $\mathcal{F}_3$ already resolves ex:periodicOrbit3_F3

examples4.tex:435 and 917 *(affects the $\cF_3$ vs $\cF_4$ narrative)*
ex:periodicOrbit3_F3 reports a 19-cell component with 6 double-edge pairs and calls the fixed-point-plus-periodic-orbit reading a conjecture; sec:dynamics_F4 then presents $\mathcal{F}_4$ as resolving it into a clean 12-cell cycle with $CH_0 \cong CH_1 \cong \mathbb{Z}_2$.
Under the live def:active_regulation, $\mathcal{F}_3$ *already* gives that 12-cell cycle, with no double edges and the same index. Attributable to that one definition change: no other divergence alters the result.
Fix: the contrast drawn between $\mathcal{F}_3$ and $\mathcal{F}_4$ at this parameter node needs restating, or a different node chosen to illustrate it.

M5 — ex:ramp_van_der_pol is labelled $\mathcal{F}_3$ but the cycle rule never fires

examples4.tex:841 *(scope)*
Neither directly specified ramp system has a semi-opaque cell with a nontrivial regulation cycle. Conditions 3.1 and 3.2 are therefore vacuous and $\mathcal{F}_3 = \mathcal{F}_2$ for both systems (verified: identical edge sets).
The computation is correct; the label overstates which rule is doing the work. Cycle cells do occur for DSGRN parameters — N3_B node 2,472,287 has one 3-cycle, N3_A node 52,718,681,992 has twelve 2-cycles and one 3-cycle.

M6 — The outer sentinel threshold differs between two chapters

RampSystemsv4.tex:388 vs RectangularGeo.tex:27 *(notation)*
defn:ramp-wall-labeling sets $\theta_{m_{K+1}} = \mathrm{GB}_n$; eq:mKnjK sets $\theta_{m_{K+1}} = \mathrm{GB}_n - \frac{1}{4}$ with $h_{m_{K+1}} = \frac{1}{4}$, so that $\theta + h = \mathrm{GB}_n$. Both are self-consistent, but any formula quantified over $\mathbf{v} \in \prod\{1, \ldots, K(n)\}$ that involves $\theta_{m_{k+1}}$ — I_n, Ξ_n, and hence the $\mathcal{H}_2$ inequality — picks up whichever convention its chapter uses, and the two disagree at the outermost cells.
Fix: one sentence saying which convention is meant where.

M7 — prop:smallh states $i \in \{0, 1, 3\}$ but proves $i = 2$ as well

RampSystemsv4.tex:624 *(statement/proof mismatch)*
The statement asserts nonemptiness for $i \in \{0, 1, 3\}$; the proof also treats $i = 2$. Separately, the constructive $\mathcal{H}_1$ bound it produces (0.02 for both published systems) is not small enough for $\mathcal{H}_2$: the largest uniform width satisfying $\mathcal{H}_0 + \mathcal{H}_1 + \mathcal{H}_2$ is 0.00517. That is consistent with the recorded concern that the constant c in the $i = 2$ branch is built from L/U, which themselves depend on h.

M8 — Two labelling issues in the trivial-index examples

examples4.tex:786 and 820 *(labelling)*
fig:trivial_index_3D_F2 is named F2 for an $\mathcal{F}_3$ example (already recorded in the knowledge base).
For the six-dimensional EMT example the eight trivial-index nodes reproduce in *count* (8 of 25) but not in the displayed labels $\{15, \ldots, 18, 21, \ldots, 24\}$. Morse-node numbering in MorseGraph breaks rank ties by the internal SCC grading value and is not canonical, so the displayed set is

an artefact of the run rather than a property of the dynamics.
The same effect shows up in the six-dimensional semiconjugacy example: the two readings give Morse graphs that differ in two edges, but every Morse set is the same collection of cells and the edges agree after exchanging the numbers of two nodes carrying the same Conley index. Any comparison of Morse graphs has to go through the cell sets, which is what `rookfields.pipeline.compare_results` does.
Consider reporting the count and the index data instead of the labels.

M9 — The divergences reach dimension six without changing any reported invariant

examples4.tex:663 and 820 (the two EMT examples) *(invariants unchanged, cells change)*
Comparing the two readings across all seventeen examples, and comparing Morse graphs through their cell sets rather than their node numbers:

- `ex:periodicOrbit3_F3` is the only example whose node count or Conley indices change (M4);
- `ex:trivial_index_3D_F3`, `ex:3d_example_1` and the second EMT example keep the same number of Morse nodes and the same multiset of Conley indices, but move cells between components. For the EMT node 1,739,757,491,101 every large component shrinks (2436 → 2394, 440 → 340, 415 → 336, 390 → 301, 300 → 262, 77 → 59) while the count of trivial-index nodes stays at 8 of 25;
- the remaining thirteen are identical.

So the corrections propagate into dimension six, but outside `ex:periodicOrbit3_F3` they do not disturb any homological quantity the text reports. The only *printed* number affected is the cell count in `ex:trivial_index_3D_F3` (M1).

Code

C1 — `active_regulation_map` implements the retired definition

CubicalBlowupGraph.py:298--309 *(changes published results)*
The target ranges over `range(self.dim)`, which is `def:active_regulation-old`. The live `def:active_regulation` (RookFields6.tex:738) requires $n, m \in J_i(\xi)$ and types the map $o_\xi : \mathrm{Act}(\xi) \to J_i(\xi)$.
Entries whose target is an *essential* direction reach the `level < 4` guard (`:461--467`) and `semi_opaque_cell` (`:470`), so $\mathcal{F}_2$ and $\mathcal{F}_3$ change. This is what makes `ex:trivial_index_3D_F3` fail to reproduce (M1) and what changes the $\mathcal{F}_3/\mathcal{F}_4$ story (M4). All two-dimensional examples are unaffected.

C2 — `decision_wall` reads the wall label off a wrapped cell

CubicalBlowupGraph.py:427--434 *(silent wrong answer)*
`defn:back-walls` shifts by $\hat{\mathbf{v}} = \sum_{n \in J_i(\xi')} \frac{1+r_n}{2} \mathbf{0}^{(n)}$, which can leave $\mathcal{X}$ at an outer endpoint. The code applies the shift with no bounds check and passes negative coordinates to `cell_index`, which wraps to an unrelated cell. Measured on decision walls that are actually built: 22%

(repressilator), 15% (3-node), 37% (4-node) have a negative coordinate, and in every one of those cases the resulting index is a different cell than intended.
Discarding those pairs changes the state transition graph broadly but changes *no* Morse-node count and *no* Conley index in any family sampled. The manuscript leaves the intended response open (safe condition $1 \le v'_j \le K(j)$); the code should at least not read a wrapped cell.

C3 — `WallLabelling.py`'s outer sentinel is $\theta + 10h$, not GB_n

`WallLabelling.py:44` *(breaks a standing assumption)*
`defn:ramp-wall-labeling` specifies $\theta_{m_{K(n)+1}} = \mathrm{GB}_n$. The code uses $\theta_{m_K} + 10\,h_{m_K}$, which depends on h. Consequences:

- it contradicts `prop:wall-labeling-const` (the labelling is constant in h on $\mathcal{H}_0$): shrinking h from 0.15 to 0.0198 flips 2 of 36 top-cell labels for the van der Pol system;
- it **breaks strong dissipativity**, the standing assumption from `CombinatorialDynamics.tex:4` onward, once $h \le 0.05$.

With the GB_n sentinel the labelling is byte-identical at every width tested. The bug is invisible at the published widths, which is presumably why it survived — but small h is exactly the regime $\mathcal{H}_1$ forces (M3).

C4 — `semi_opaque_cell` omits the two excluded classes

`CubicalBlowupGraph.py:470--475` *(definition not enforced)*
`defn:partially_opaque` requires $\xi \in \mathcal{X} \setminus (\mathcal{X}^{(N)} \cup \mathrm{bdy}(\mathcal{X}))$ in addition to o_ξ being a bijection; the code checks only the bijection. A top cell has an empty regulation map and passes vacuously. Enforcing the exclusions changes the state transition graph on 16/60 and 20/80 parameters at levels 3 and 4, with no change to Morse nodes or Conley indices.

C5 — `IsomorphismQuery` is broken against current HEAD

`IsomorphismQuery.py:23` *(crashes)*
`DSGRN.isomorphic_morse_graphs` calls `.split(':')` on a vertex label, but commit `fb10673` changed `MorseGraph.vertex_label` to return a list — so this raises `AttributeError`. The loop also uses `continue` where `break` is meant, so a parameter matching two representatives is added to both isomorphism classes.
Unrelated to the manuscript, but it means the query is unusable as shipped.

Example by example

Each block lists what the text claims and what was computed. A row is shaded red when the computed value differs from the printed one.

ex:saddle_saddle_3D_intro

introduction6.tex:402 · *network N3_A, parameter node* 52,718,681,992 · $\mathcal{F}_3$

Text claims: 25 Morse nodes, every Conley index nonzero (hyperbolic fixed points)

spec = legacy 0.09 s

Quantity	Claimed	Computed	
number of Morse nodes	25	25	**ok**
every Morse node has nonzero index	[]	[]	**ok**

spec = paper 0.19 s

Quantity	Claimed	Computed	
number of Morse nodes	25	25	**ok**
every Morse node has nonzero index	[]	[]	**ok**

Computed: 25 Morse nodes, index set `[4, 4, 4]`, 1000 STG vertices and 3644 edges.

Same parameter node as ex:saddle_saddle_3D_example.

ex:introPeriodic

introduction6.tex:498 · *ramp system intro_periodic* · $\mathcal{F}_3$

Text claims: Morse node 0 has CH_0 = CH_1 = Z_2 (stable periodic orbit index)

spec = legacy 0.09 s

Quantity	Claimed	Computed	
CH_*(0) is the stable-periodic-orbit signature	[1, 1, 0, 0]	[1, 1, 0, 0]	**ok**

spec = paper 0.38 s

Quantity	Claimed	Computed	
CH_*(0) is the stable-periodic-orbit signature	[1, 1, 0, 0]	[1, 1, 0, 0]	**ok**

Computed: 2 Morse nodes, index set `[8, 6, 3]`, 2016 STG vertices and 7025 edges.

ex:saddlesaddlebif

examples4.tex:39 · *network N2_a, parameter node* 752 · $\mathcal{F}_3$

Text claims: two admissible connection matrices; 1600 nodes in about 8 s

No numeric claim in the text to check.

Computed: 9 Morse nodes, index set `[3, 3]`, 64 STG vertices and 153 edges.

The 8 s is for all 1600 parameter nodes, not this one.

Runtime: 0.00 s here against 8 s reported.

ex:saddle_saddle_3D_example

examples4.tex:263 · *network N3_A, parameter node* 52,718,681,992 · $\mathcal{F}_3$

Text claims: 25 equilibrium-type Morse nodes; 4096 connection matrices; 0.19 s

spec = legacy 0.09 s

Quantity	Claimed	Computed	
number of Morse nodes	25	25	**ok**
every Morse node has nonzero index	[]	[]	**ok**

spec = paper 0.18 s

Quantity	Claimed	Computed	
number of Morse nodes	25	25	**ok**
every Morse node has nonzero index	[]	[]	**ok**

Computed: 25 Morse nodes, index set `[4, 4, 4]`, 1000 STG vertices and 3644 edges.

Runtime: 0.09 s here against 0.19 s reported.

ex:periodicOrbit3

examples4.tex:307 · *ramp system intro_periodic* · $\mathcal{F}_3$

Text claims: I = {0..8}x{0..6}x{0..3}; M(0) has the periodic-orbit index; 0.32 s

spec = legacy 0.09 s

Quantity	Claimed	Computed	
index set I (top index per coordinate)	[8, 6, 3]	[8, 6, 3]	**ok**
CH_*(0) is the stable-periodic-orbit signature	[1, 1, 0, 0]	[1, 1, 0, 0]	**ok**

spec = paper 0.39 s

Quantity	Claimed	Computed	
index set I (top index per coordinate)	[8, 6, 3]	[8, 6, 3]	**ok**

Quantity	Claimed	Computed	
CH_*(0) is the stable-periodic-orbit signature	[1, 1, 0, 0]	[1, 1, 0, 0]	ok

Computed: 2 Morse nodes, index set [8, 6, 3], 2016 STG vertices and 7025 edges.

Runtime: 0.09 s here against 0.32 s reported.

ex:periodicOrbit3_F3

examples4.tex:435 · *network N3_B, parameter node* 2,472,287 · $\mathcal{F}_3$

Text claims: Morse node 1 has 19 cells, 6 double-edge pairs, and an equilibrium cell

spec = legacy 0.04 s

Quantity	Claimed	Computed	
cells in the multi-cell component (node 1)	19	19	ok
double-edge pairs in the multi-cell component (node 1)	6	6	ok

spec = paper 0.10 s

Quantity	Claimed	Computed	
cells in the multi-cell component (node 1)	19	12	differs
double-edge pairs in the multi-cell component (node 1)	6	0	differs

Computed: 13 Morse nodes, index set [3, 4, 3], 640 STG vertices and 2377 edges.

The source calls the fixed-point-plus-periodic-orbit reading a conjecture. Under the 'paper' spec the component is instead a clean 12-cell cycle with no double edges and the stable-periodic-orbit index – i.e. F_3 already yields the resolution the text attributes to F_4. Attributable to D1 alone.

ex:mccord

examples4.tex:489 · *network N2_b, parameter node* 974 · $\mathcal{F}_3$

Text claims: 3 Morse nodes with order 0¡2, 1¡2; CH_0(0)=CH_0(1)=Z_2, CH_1(2)=Z_2

spec = legacy 0.00 s

Quantity	Claimed	Computed	
number of Morse nodes	3	3	ok
Morse graph edges	[[2, 0], [2, 1]]	[[2, 0], [2, 1]]	ok
Conley index CH_*(0)	[1, 0, 0]	[1, 0, 0]	ok
Conley index CH_*(1)	[1, 0, 0]	[1, 0, 0]	ok
Conley index CH_*(2)	[0, 1, 0]	[0, 1, 0]	ok

spec = paper 0.00 s

Quantity	Claimed	Computed	
number of Morse nodes	3	3	ok
Morse graph edges	[[2, 0], [2, 1]]	[[2, 0], [2, 1]]	ok
Conley index CH_*(0)	[1, 0, 0]	[1, 0, 0]	ok
Conley index CH_*(1)	[1, 0, 0]	[1, 0, 0]	ok
Conley index CH_*(2)	[0, 1, 0]	[0, 1, 0]	ok

Computed: 3 Morse nodes, index set [3, 3], 64 STG vertices and 149 edges.

R(974) is the running wall labelling of fig:wall_labeling(A).

ex:3d_example_1

examples4.tex:538 · *network N3_B, parameter node* 2,472,286 · $\mathcal{F}_3$

Text claims: 13-node Morse representation, unique connection matrix; 0.14 s

spec = legacy 0.04 s

Quantity	Claimed	Computed	
number of Morse nodes	13	13	ok
every Morse node has nonzero index	[]	[]	ok

spec = paper 0.10 s

Quantity	Claimed	Computed	
number of Morse nodes	13	13	ok
every Morse node has nonzero index	[]	[]	ok

Computed: 13 Morse nodes, index set [3, 4, 3], 640 STG vertices and 2378 edges.

Runtime: 0.04 s here against 0.14 s reported.

EMT-6D-semiconjugacy

examples4.tex:663 · *network N6_EMT, parameter node* 2,684,686,006 · $\mathcal{F}_2$

Text claims: Morse graph and connection matrix in about 2.2 minutes

No numeric claim in the text to check.

Computed: 9 Morse nodes, index set [2, 3, 5, 3, 4, 2], 276480 STG vertices and 2501559 edges.

Runtime: 88.64 s here against 132 s reported.

ex:trivial_index_2D

examples4.tex:766 · *network N2_a, parameter node* 47 · $\mathcal{F}_3$

Text claims: 4 Morse nodes: three with nonzero index, one trivial

spec = legacy 0.00 s

Quantity	Claimed	Computed	
number of Morse nodes	4	4	ok
trivial-index Morse nodes	1	1	ok

spec = paper 0.00 s

Quantity	Claimed	Computed	
number of Morse nodes	4	4	ok
trivial-index Morse nodes	1	1	ok

Computed: 4 Morse nodes, index set [3, 3], 64 STG vertices and 147 edges.

ex:trivial_index_3D_F3

examples4.tex:786 · *network N3_A, parameter node* 65,571,607,721 · $\mathcal{F}_3$

Text claims: three one-cell equilibrium nodes plus a 61-cell, 12-double-edge trivial node

spec = legacy 0.06 s

Quantity	Claimed	Computed	
number of Morse nodes	4	4	ok
trivial-index Morse nodes	1	1	ok
cells in the trivial-index component (node 3)	61	75	differs
double-edge pairs in the trivial-index component (node 3)	12	16	differs

spec = paper 0.17 s

Quantity	Claimed	Computed	
number of Morse nodes	4	4	ok
trivial-index Morse nodes	1	1	ok
cells in the trivial-index component (node 2)	61	61	ok
double-edge pairs in the trivial-index component (node 2)	12	12	ok

Computed: 4 Morse nodes, index set [4, 4, 4], 1000 STG vertices and 3589 edges.

fig:trivial_index_3D_F2 is misnamed; the example uses F_3. The published 61/12 reproduce under the 'paper' spec but not under 'legacy', which gives 75 cells and 16 double-edge pairs.

ex:trivial_indices

examples4.tex:801 · *network N3_A, parameter node* 52,717,613,010 · $\mathcal{F}_3$

Text claims: 8 connection matrices; trivial index at p in {10,11,13,18,27} with recurrent components of 44, 42, 42, 14, 14 cells; 0.2 s

spec = legacy 0.10 s

Quantity	Claimed	Computed	
trivial-index node set	[10, 11, 13, 18, 27]	[10, 11, 13, 18, 27]	ok
cells in Morse set 10	44	44	ok
cells in Morse set 11	42	42	ok
cells in Morse set 13	42	42	ok
cells in Morse set 18	14	14	ok
cells in Morse set 27	14	14	ok

spec = paper 0.21 s

Quantity	Claimed	Computed	
trivial-index node set	[10, 11, 13, 18, 27]	[10, 11, 13, 18, 27]	ok
cells in Morse set 10	44	44	ok
cells in Morse set 11	42	42	ok
cells in Morse set 13	42	42	ok
cells in Morse set 18	14	14	ok
cells in Morse set 27	14	14	ok

Computed: 36 Morse nodes, index set [4, 4, 4], 1000 STG vertices and 3642 edges.

Runtime: 0.10 s here against 0.2 s reported.

EMT-6D-trivial-indices

examples4.tex:820 · *network N6_EMT, parameter node* 1,739,757,491,101 · $\mathcal{F}_2$

Text claims: eight trivial-index nodes, p in {15..18, 21..24}; about 2.5 minutes

spec = legacy 147.79 s

Quantity	Claimed	Computed	
number of Morse nodes	25	25	ok
trivial-index Morse nodes	8	8	ok

spec = paper 171.05 s

Quantity	Claimed	Computed	
number of Morse nodes	25	25	ok
trivial-index Morse nodes	8	8	ok

Computed: 25 Morse nodes, index set [2, 3, 5, 3, 4, 2], 276480 STG vertices and 2507717 edges.

*The count reproduces exactly (8 of 25). The displayed node *labels* do not: legacy gives {15,17,18,19,21,22,23,24} and paper gives {15,16,18,19,21,22,23,24}. Morse-node numbering in 'MorseGraph' (MorseGraph.py:79-86) sorts by rank and breaks ties by the internal SCC grading value, so it is not canonical; only the count and the index data are.*

Runtime: 147.79 s here against 150 s reported.

ex:ramp_van_der_pol

examples4.tex:841 · *ramp system van_der_pol* · $\mathcal{F}_3$

Text claims: F_3 and Morse graph for a multi-ramp planar system; 0.03 s

spec = legacy 0.01 s

Quantity	Claimed	Computed	
index set I (top index per coordinate)	[6, 6]	[6, 6]	ok

spec = paper 0.02 s

Quantity	Claimed	Computed	
index set I (top index per coordinate)	[6, 6]	[6, 6]	ok

Computed: 2 Morse nodes, index set [6, 6], 196 STG vertices and 421 edges.

The example does not verify h in H_3; see the hypotheses report.

Runtime: 0.01 s here against 0.03 s reported.

ex:periodicOrbit3_F4

examples4.tex:917 · *network N3_B, parameter node* 2,472,287 · $\mathcal{F}_4$

Text claims: I = {0..4}x{0..4}x{0..3}; M(1) has the periodic-orbit index; 12-cell cycle with no double edges; 0.12 s

spec = legacy 0.04 s

Quantity	Claimed	Computed	
index set I (top index per coordinate)	[4, 4, 3]	[3, 4, 3]	differs
cells in the periodic-orbit component (node 1)	12	12	ok
double-edge pairs in the periodic-orbit component (node 1)	0	0	ok

spec = paper 0.10 s

Quantity	Claimed	Computed	
index set I (top index per coordinate)	[4, 4, 3]	[3, 4, 3]	differs
cells in the periodic-orbit component (node 1)	12	12	ok
double-edge pairs in the periodic-orbit component (node 1)	0	0	ok

Computed: 14 Morse nodes, index set [3, 4, 3], 640 STG vertices and 2370 edges.

No ODE realisation is claimed for any F_4 example (examples4.tex:915). The displayed index set is off in its first factor: node 1 of N3_B has two out-edges, so K(1) = 2 and I = {0..3} x {0..4} x {0..3}. Everything else in the example reproduces exactly.

Runtime: 0.04 s here against 0.12 s reported.

ex:periodicOrbit5

examples4.tex:979 · *network N5, parameter node* 5,103,162,287 · $\mathcal{F}_4$

Text claims: 7 equilibrium cells (Morse nodes 1-7); node 0 is a 12-cell cycle with the periodic-orbit index; 9.15 s

spec = legacy 4.91 s

Quantity	Claimed	Computed	
number of Morse nodes	8	8	**ok**
cells in the periodic-orbit component (node 0)	12	12	**ok**
equilibrium cells in X	7	7	**ok**

spec = paper 19.42 s

Quantity	Claimed	Computed	
number of Morse nodes	8	8	**ok**
cells in the periodic-orbit component (node 0)	12	12	**ok**
equilibrium cells in X	7	7	**ok**

Computed: 8 Morse nodes, index set `[3, 5, 4, 2, 2]`, 34560 STG vertices and 246527 edges.

Runtime: 4.91 s here against 9.15 s reported.

ex:missing_periodic_3D_F4

examples4.tex:1063 · *network N3_B, parameter node* 2,023,186 · $\mathcal{F}_4$

Text claims: Morse node 3 is a 29-cell SCC with 4 unresolved double-edge pairs

spec = legacy 0.04 s

Quantity	Claimed	Computed	
cells in the multi-cell component (node 3)	29	29	**ok**
double-edge pairs in the multi-cell component (node 3)	4	4	**ok**

spec = paper 0.10 s

Quantity	Claimed	Computed	
cells in the multi-cell component (node 3)	29	29	**ok**
double-edge pairs in the multi-cell component (node 3)	4	4	**ok**

Computed: 9 Morse nodes, index set `[3, 4, 3]`, 640 STG vertices and 2374 edges.

The 29-cell component carries a fixed-point index (1,0,0,0), not a trivial one; the source conjectures its invariant part holds both a fixed point and a periodic orbit.

Supporting evidence

Which definition change affects what

Parameters whose result changes relative to **legacy**, per divergence, sampled over whole parameter graphs where feasible.

Network	Divergence	Level	STG	Morse nodes	Conley
repressilator	D1-regulation-codomain	2	7	1	1
repressilator	D2-back-wall-strict	2	19	0	0
repressilator	D2-back-wall-shifted	2	19	0	0
repressilator	D1-regulation-codomain	3	7	1	1
repressilator	D2-back-wall-strict	3	19	0	0
repressilator	D2-back-wall-shifted	3	19	0	0
repressilator	D2-back-wall-strict	4	19	0	0
repressilator	D2-back-wall-shifted	4	19	0	0
cycle3	D1-regulation-codomain	2	38	1	1
cycle3	D2-back-wall-strict	2	112	0	0
cycle3	D2-back-wall-shifted	2	112	0	0
cycle3	D1-regulation-codomain	3	38	1	1
cycle3	D2-back-wall-strict	3	112	0	0
cycle3	D2-back-wall-shifted	3	112	0	0
cycle3	D2-back-wall-strict	4	112	0	0
cycle3	D2-back-wall-shifted	4	112	0	0
cycle4	D1-regulation-codomain	2	33	4	4
cycle4	D2-back-wall-strict	2	54	0	0
cycle4	D2-back-wall-shifted	2	54	0	0
cycle4	D1-regulation-codomain	3	34	0	0
cycle4	D2-back-wall-strict	3	54	0	0
cycle4	D2-back-wall-shifted	3	54	0	0
cycle4	D4-semi-opaque-guard	3	16	0	0
cycle4	D1-regulation-codomain	4	5	0	0
cycle4	D2-back-wall-strict	4	54	0	0
cycle4	D2-back-wall-shifted	4	54	0	0
cycle4	D4-semi-opaque-guard	4	16	0	0
N3_B	D1-regulation-codomain	2	50	2	2
N3_B	D2-back-wall-strict	2	80	0	0
N3_B	D2-back-wall-shifted	2	80	0	0
N3_B	D1-regulation-codomain	3	50	2	2
N3_B	D2-back-wall-strict	3	80	0	0
N3_B	D2-back-wall-shifted	3	80	0	0
N3_B	D4-semi-opaque-guard	3	20	0	0
N3_B	D2-back-wall-strict	4	80	0	0
N3_B	D2-back-wall-shifted	4	80	0	0
N3_B	D4-semi-opaque-guard	4	20	0	0

Three further readings were parameterised and changed *nothing* anywhere: evaluating every back-wall choice (positive evidence for `prop:back_wall_well_defined`); computing $\mathcal{F}_3$ as an intersection plus an unconditional union rather than as a cascade; and widening the domain of o_ξ.

Analytic admissibility of the ramp parameters

System	half-width	$\mathcal{H}_0$	$\Lambda(R)$	$\mathcal{H}_1$	$\mathcal{H}_2$	$\mathcal{H}_3$
van_der_pol	published	ok	ok	differs	differs	differs
van_der_pol	0.05	ok	ok	differs	differs	differs
van_der_pol	0.02	ok	ok	differs	differs	differs
van_der_pol	0.01	ok	ok	ok	differs	ok
van_der_pol	0.005	ok	ok	ok	ok	ok
van_der_pol	0.002	ok	ok	ok	ok	ok
van_der_pol	0.001	ok	ok	ok	ok	ok
intro_periodic	published	ok	ok	differs	differs	differs
intro_periodic	0.05	ok	ok	differs	differs	differs
intro_periodic	0.02	ok	ok	differs	differs	differs
intro_periodic	0.01	ok	ok	ok	differs	ok
intro_periodic	0.005	ok	ok	ok	ok	ok
intro_periodic	0.002	ok	ok	ok	ok	ok
intro_periodic	0.001	ok	ok	ok	ok	ok

The Morse graph is identical at every width in this table, so repairing the width does not disturb any computed conclusion.

Numerical alignment of the rectangular geometrization

The inward normal component of the ramp field, sampled on every embedded wall that the combinatorial map orients. thm:R1ABlattice asserts all of them are positive for $\mathcal{F}_1$.

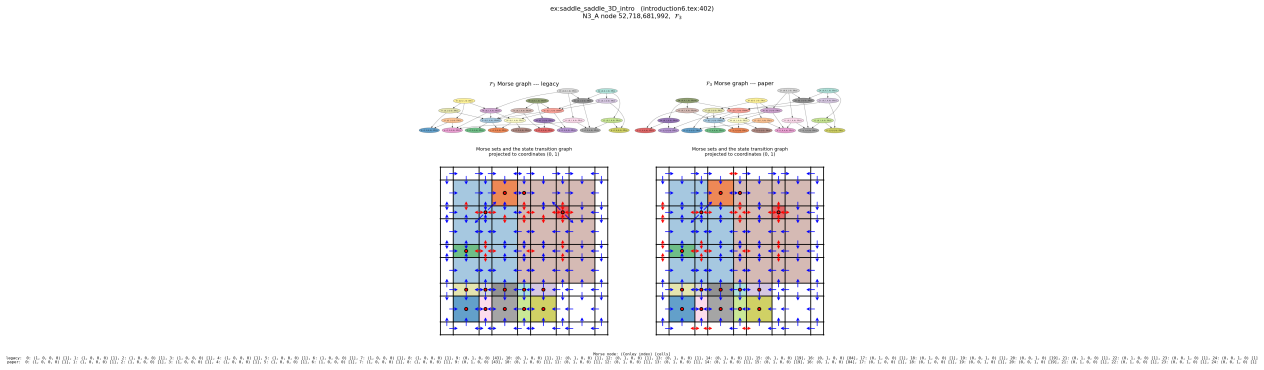
System	half-width	Level	walls	worst inward	aligned
van_der_pol	published	1	298	−0.26	differs
van_der_pol	published	2	312	−2	differs
van_der_pol	published	3	312	−2	differs
van_der_pol	0.02	1	298	−0	differs
van_der_pol	0.02	2	312	−1.74	differs
van_der_pol	0.02	3	312	−1.74	differs
van_der_pol	0.005	1	298	0.03	ok
van_der_pol	0.005	2	312	−1.71	differs
van_der_pol	0.005	3	312	−1.71	differs
van_der_pol	0.002	1	298	0.036	ok
van_der_pol	0.002	2	312	−1.7	differs
van_der_pol	0.002	3	312	−1.7	differs
intro_periodic	published	1	4099	−0.26	differs
intro_periodic	published	2	4180	−2	differs
intro_periodic	published	3	4180	−2	differs
intro_periodic	0.02	1	4099	−0	differs
intro_periodic	0.02	2	4180	−1.74	differs
intro_periodic	0.02	3	4180	−1.74	differs
intro_periodic	0.005	1	4099	0.03	ok
intro_periodic	0.005	2	4180	−1.71	differs
intro_periodic	0.005	3	4180	−1.71	differs
intro_periodic	0.002	1	4099	0.036	ok
intro_periodic	0.002	2	4180	−1.7	differs
intro_periodic	0.002	3	4180	−1.7	differs

At an admissible width $\mathcal{F}_1$ is aligned with zero failures; at the published widths it is not, so the width condition

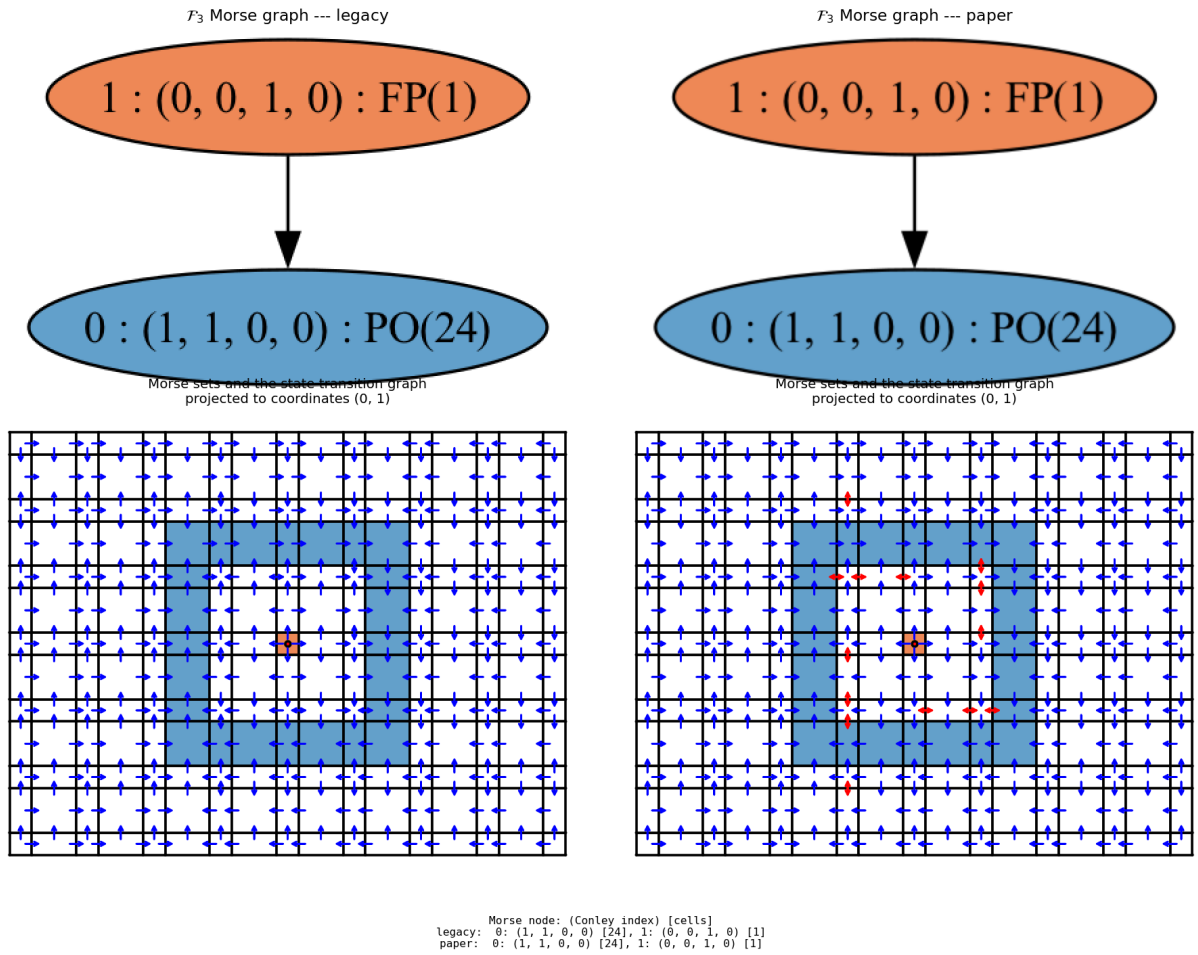
is load-bearing. Every $\mathcal{F}_2/\mathcal{F}_3$ failure sits on a wall that $\mathcal{F}_1$ left as a double edge — precisely the faces the R2 GO-manifold and R3 cycle-surface constructions are meant to replace.

Morse graphs and Morse sets

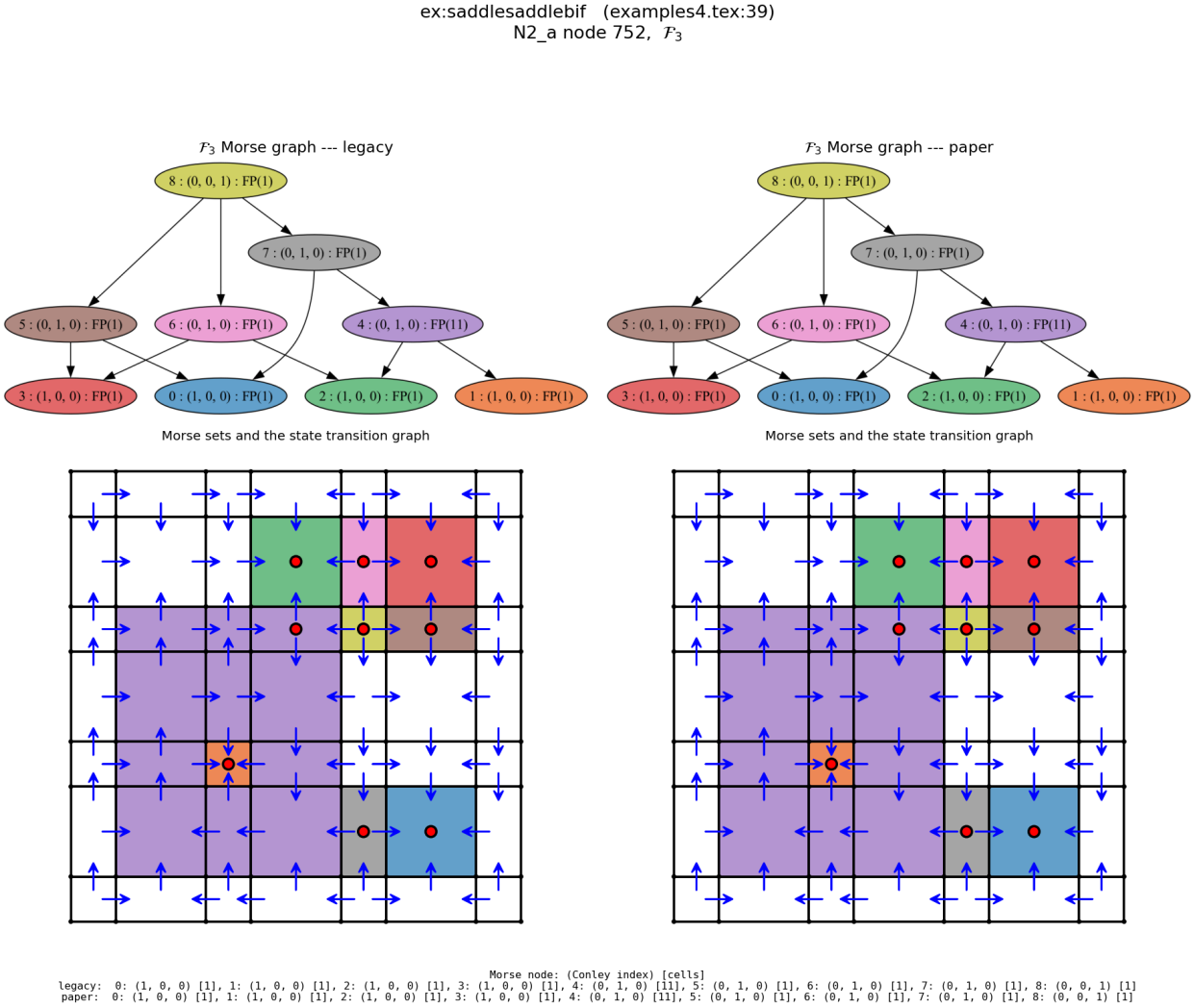
For each example, the Morse graph and the Morse sets under both specifications. Morse graph nodes carry the Conley index over $\mathbb{Z}_2$ and the classification FP (fixed point), PO (periodic orbit), T (trivial index) or M (more complex), with the number of cells in parentheses. In the Morse set panels blue arrows are single edges of the state transition graph, red arrows are double edges, and a red disc marks a self-edge; above dimension two the complex is projected onto the first two coordinates. Plots are produced by DSGRN_utils.PlotMorseGraph and PlotMorseSets, the same routines the monograph's figures use.



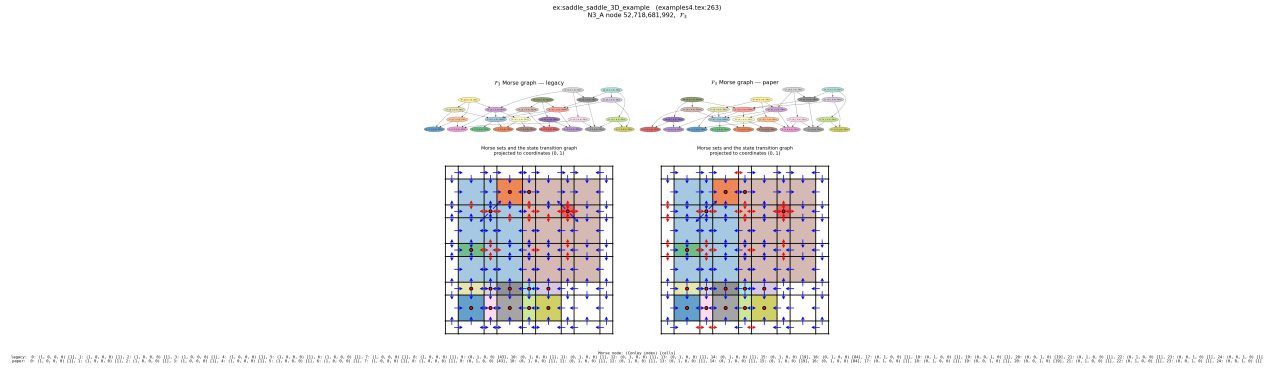
Both specifications agree. ex:saddle_saddle_3D_intro (introduction6.tex:402), $\mathcal{F}_3$, dimension 3.



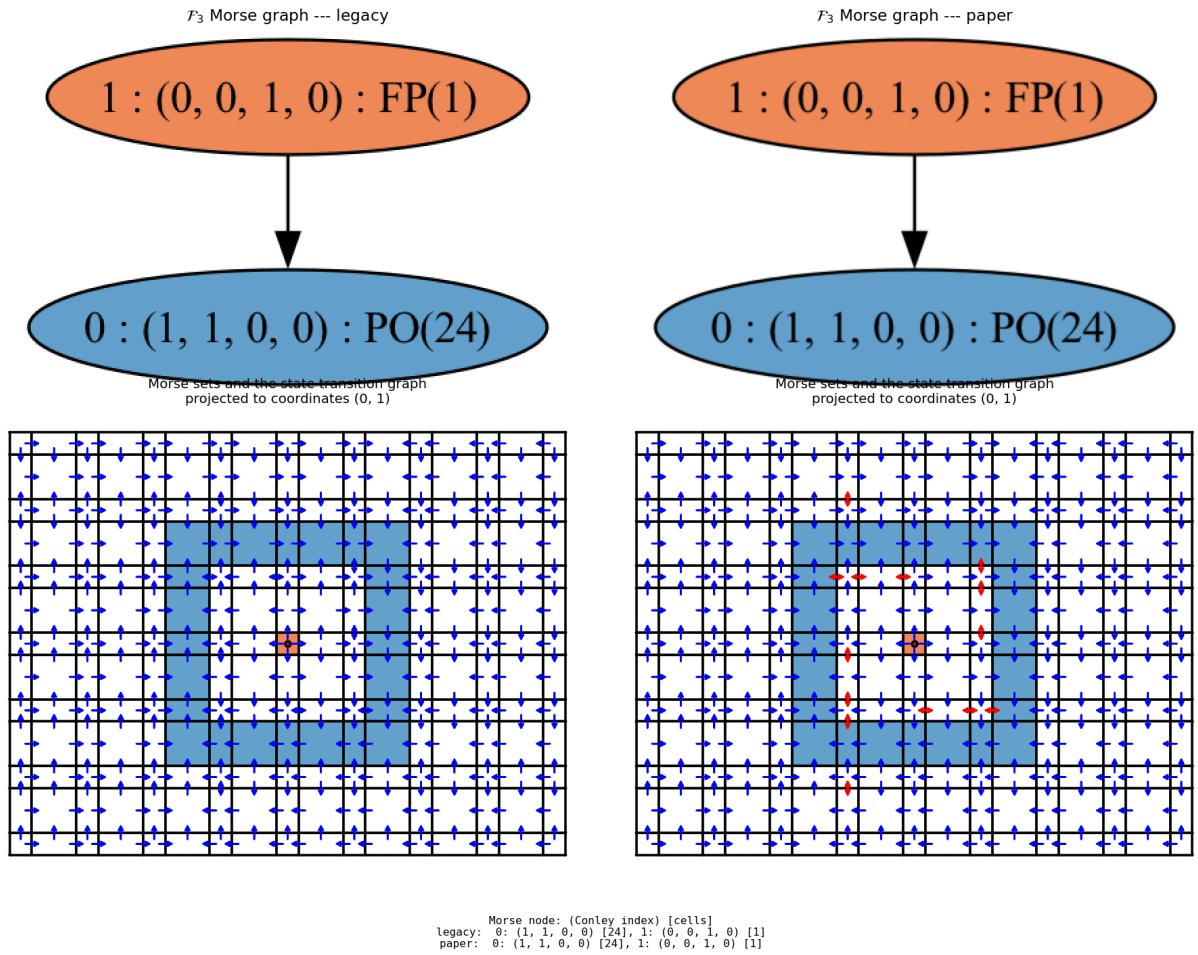
Both specifications agree. `ex:introPeriodic` (`introduction6.tex:498`), $\mathcal{F}_3$, dimension 3.



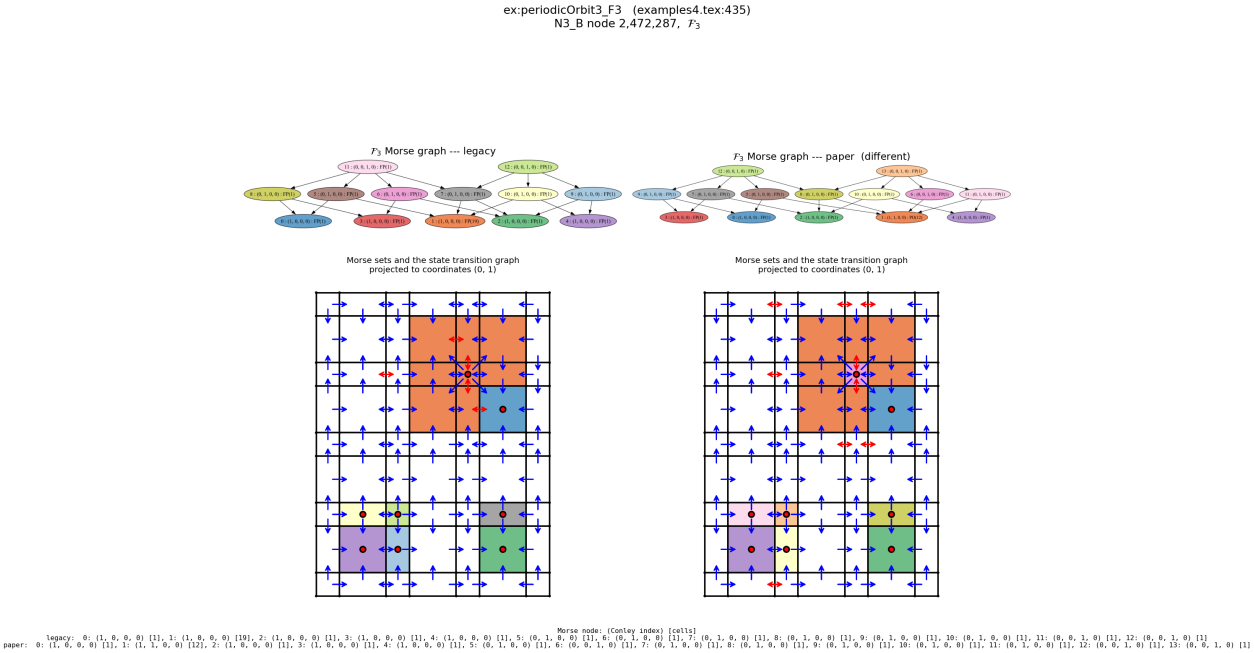
Both specifications agree. `ex:saddlesaddlebif` (`examples4.tex:39`), $\mathcal{F}_3$, dimension 2.



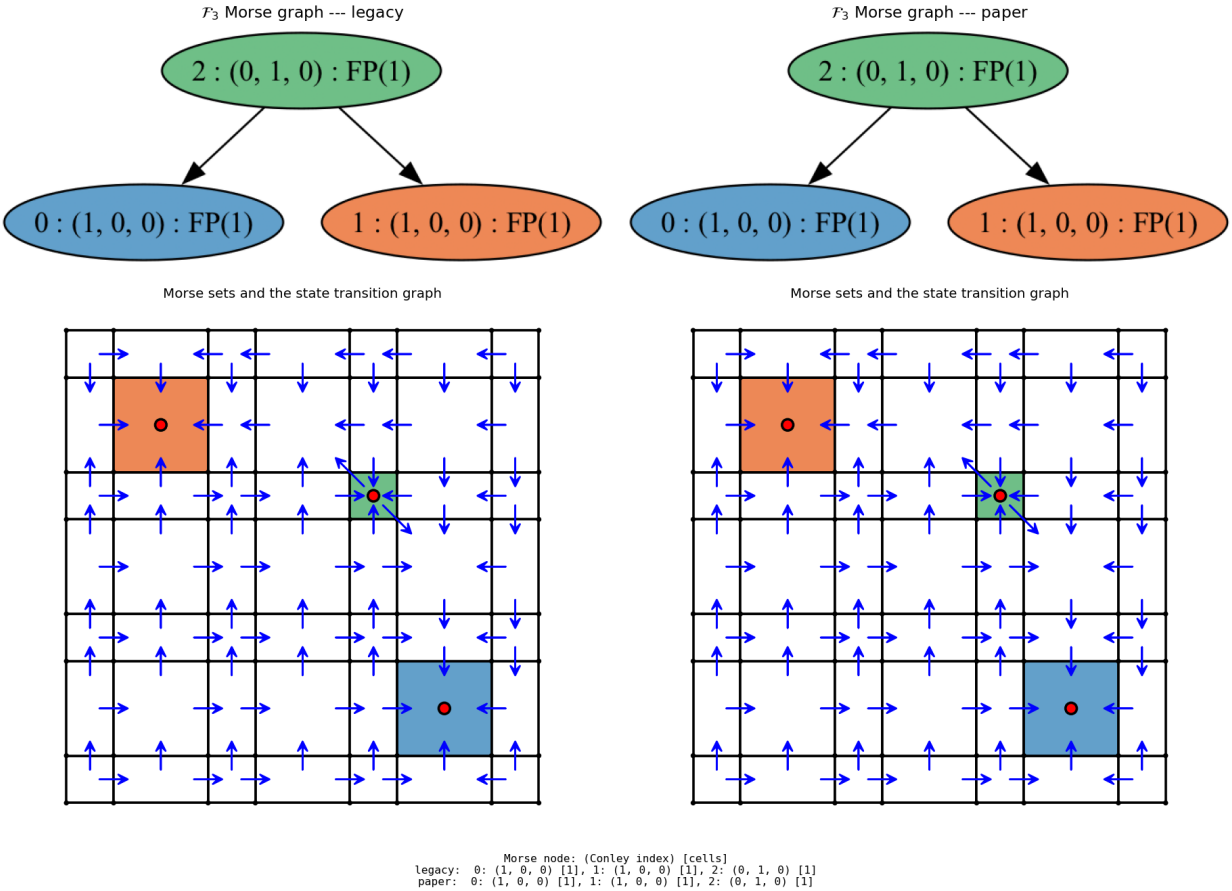
Both specifications agree. `ex:saddle_saddle_3D_example` (`examples4.tex:263`), $\mathcal{F}_3$, dimension 3.



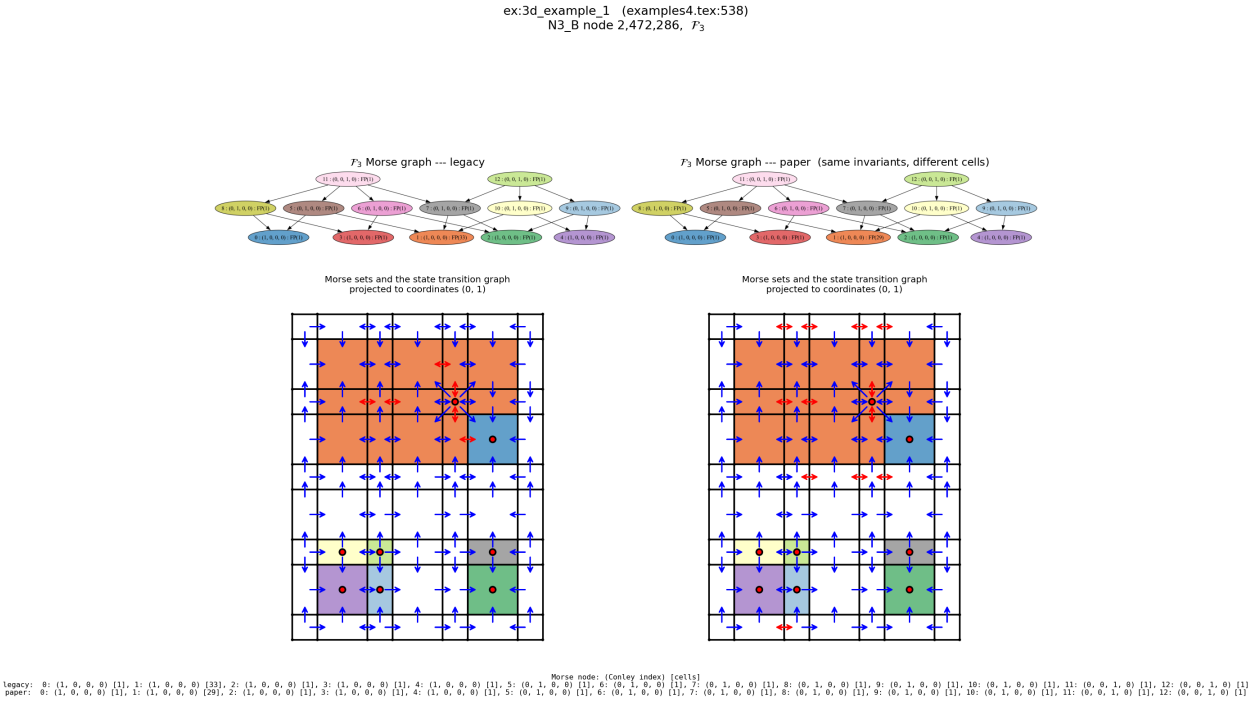
Both specifications agree. `ex:periodicOrbit3` (`examples4.tex:307`), $\mathcal{F}_3$, dimension 3.



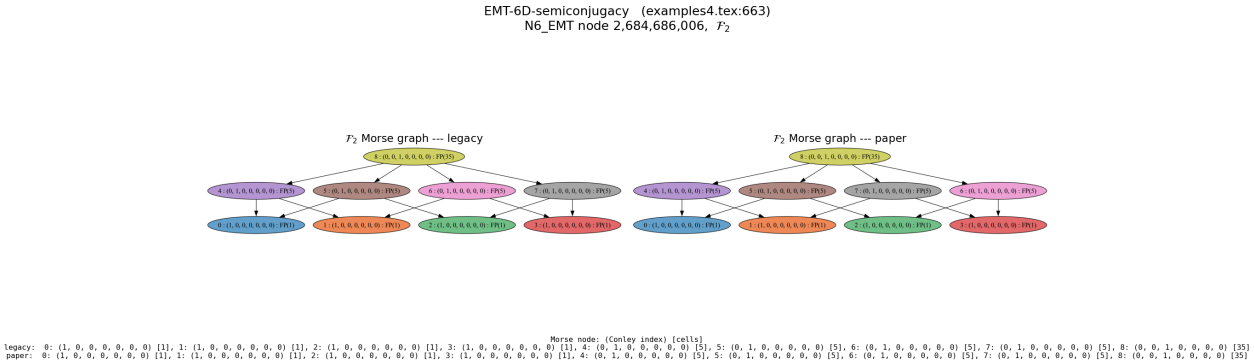
The two specifications disagree. `ex:periodicOrbit3_F3` (`examples4.tex:435`), $\mathcal{F}_3$, dimension 3.



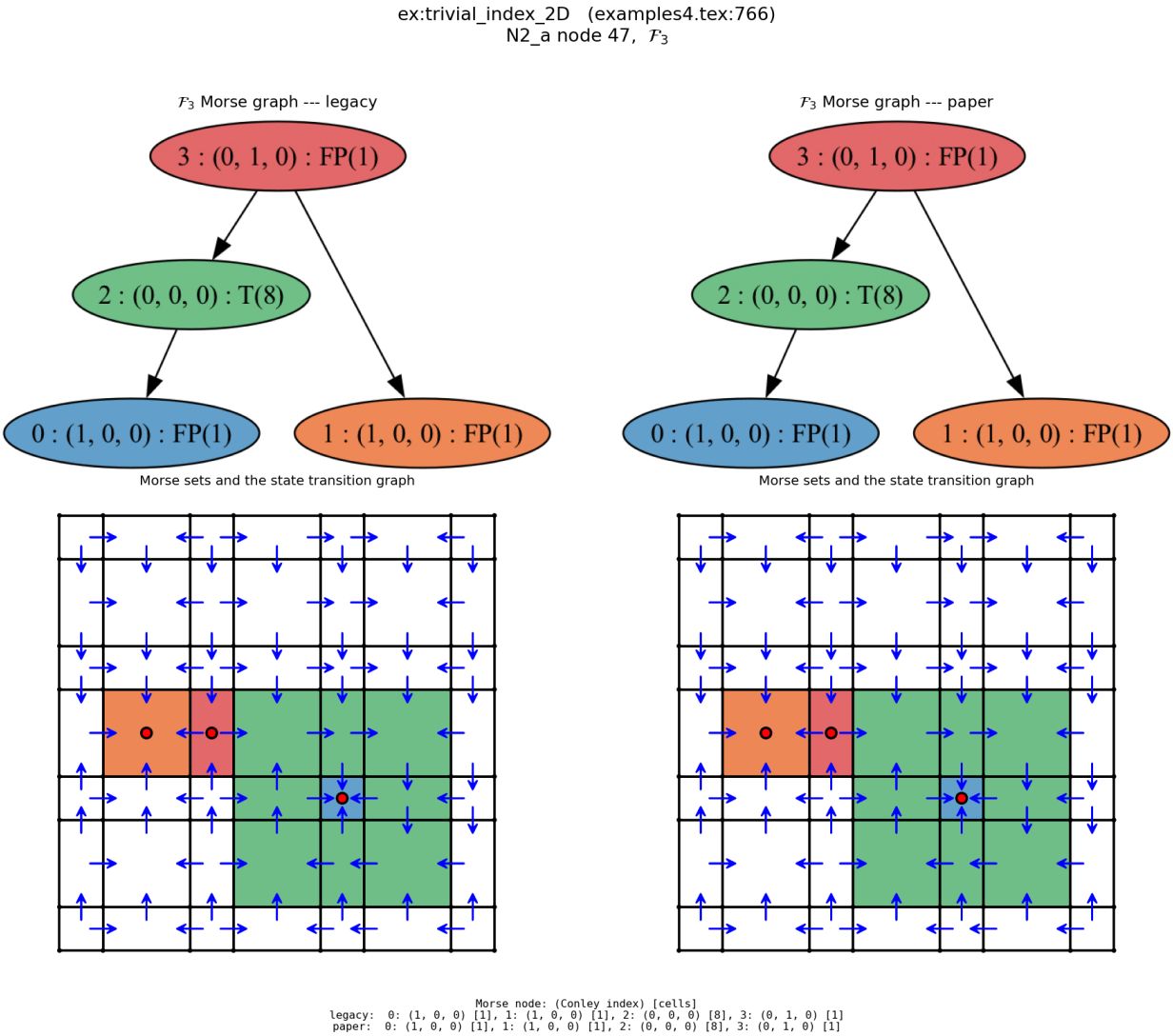
Both specifications agree. `ex:mccord` (`examples4.tex:489`), $\mathcal{F}_3$, dimension 2.



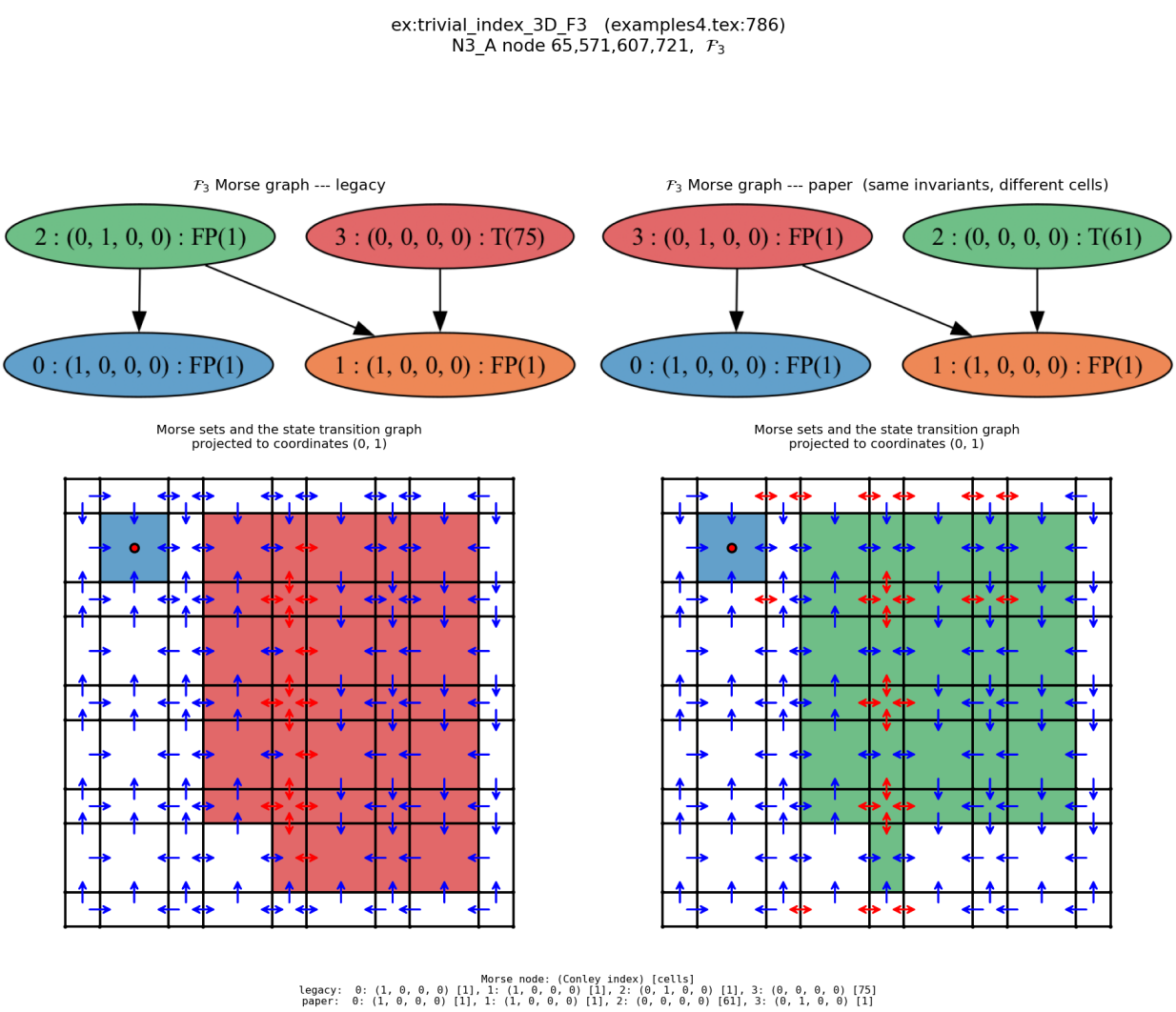
The two specifications disagree. `ex:3d_example_1` (`examples4.tex:538`), $\mathcal{F}_3$, dimension 3.



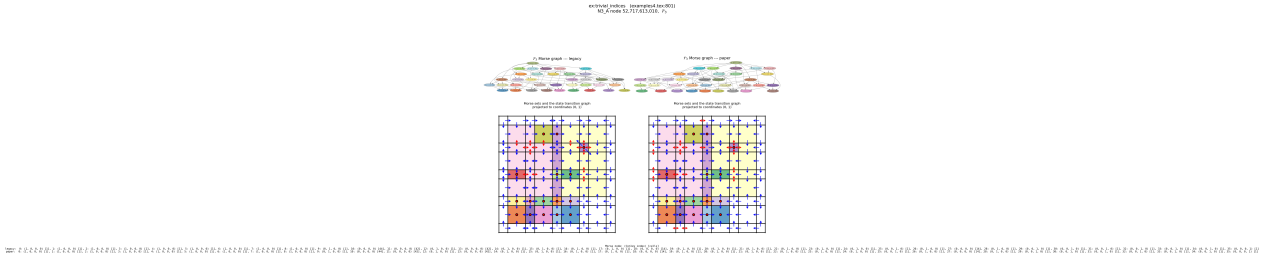
Both specifications agree. `EMT-6D-semiconjugacy` (`examples4.tex:663`), $\mathcal{F}_2$, dimension 6.



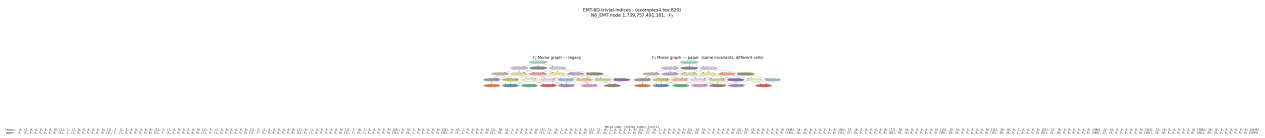
Both specifications agree. `ex:trivial_index_2D` (`examples4.tex:766`), $\mathcal{F}_3$, dimension 2.



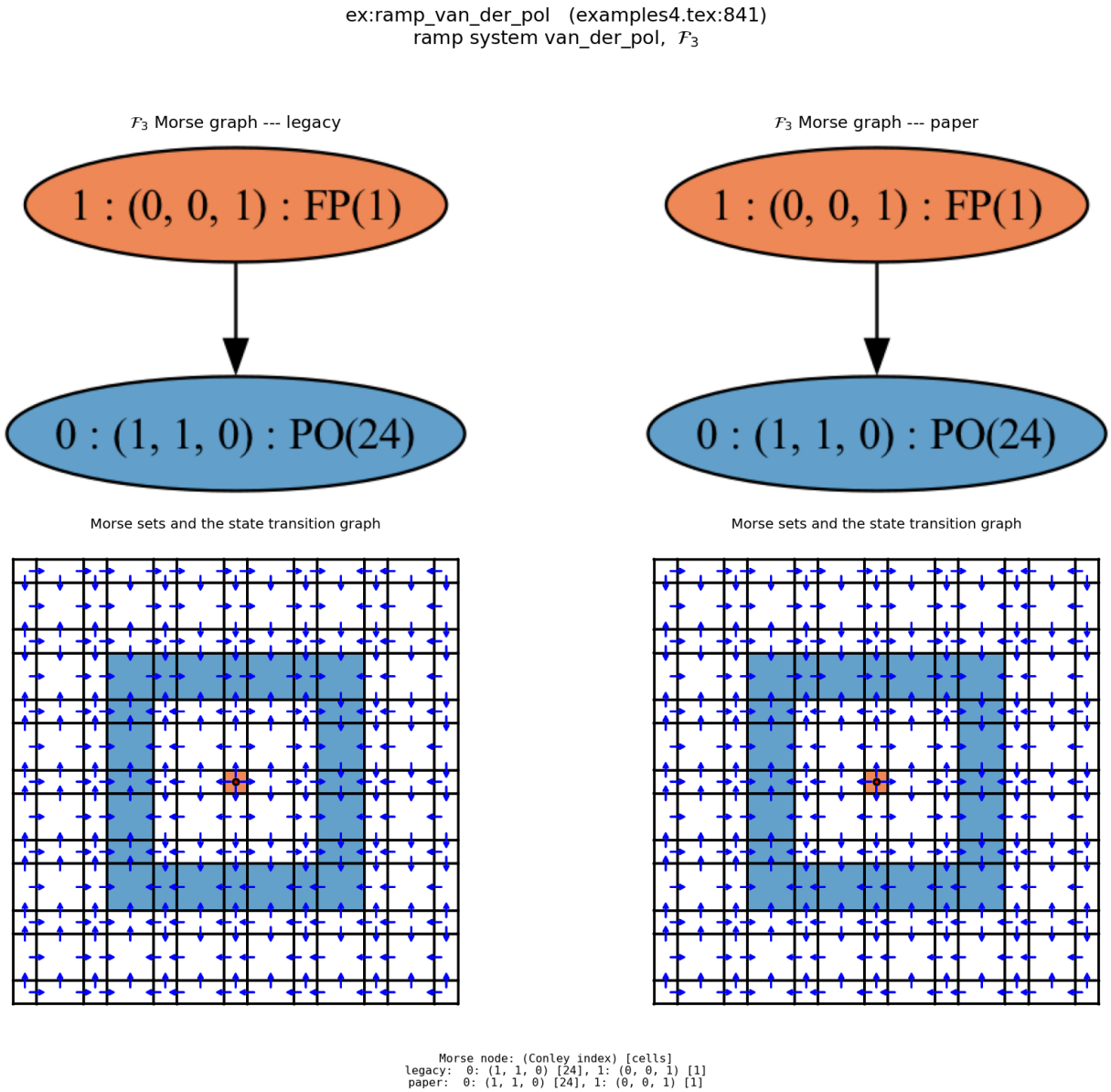
The two specifications disagree. `ex:trivial_index_3D_F3` (`examples4.tex:786`), $\mathcal{F}_3$, dimension 3.



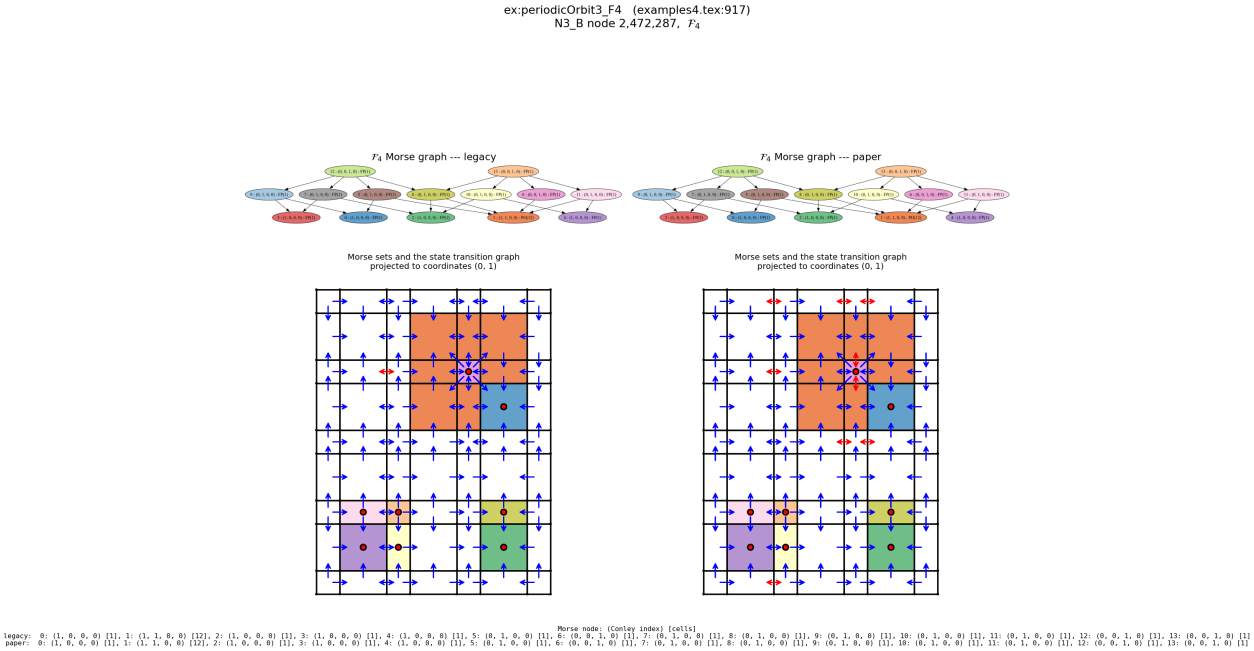
Both specifications agree. `ex:trivial_indices` (`examples4.tex:801`), $\mathcal{F}_3$, dimension 3.



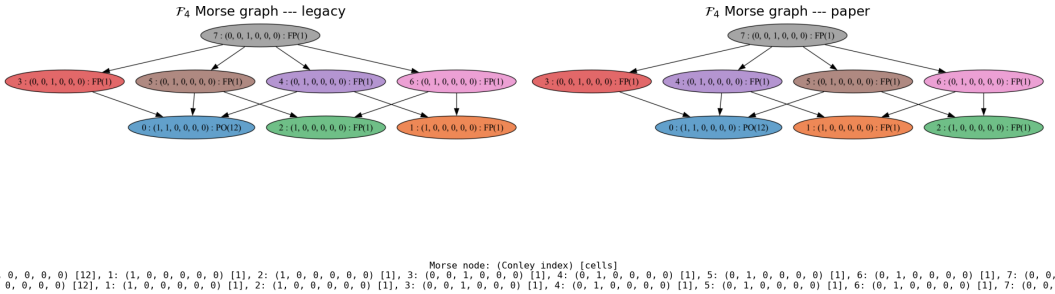
The two specifications disagree. `EMT-6D-trivial-indices` (`examples4.tex:820`), $\mathcal{F}_2$, dimension 6.



Both specifications agree. `ex:ramp_van_der_pol` (`examples4.tex:841`), $\mathcal{F}_3$, dimension 2.

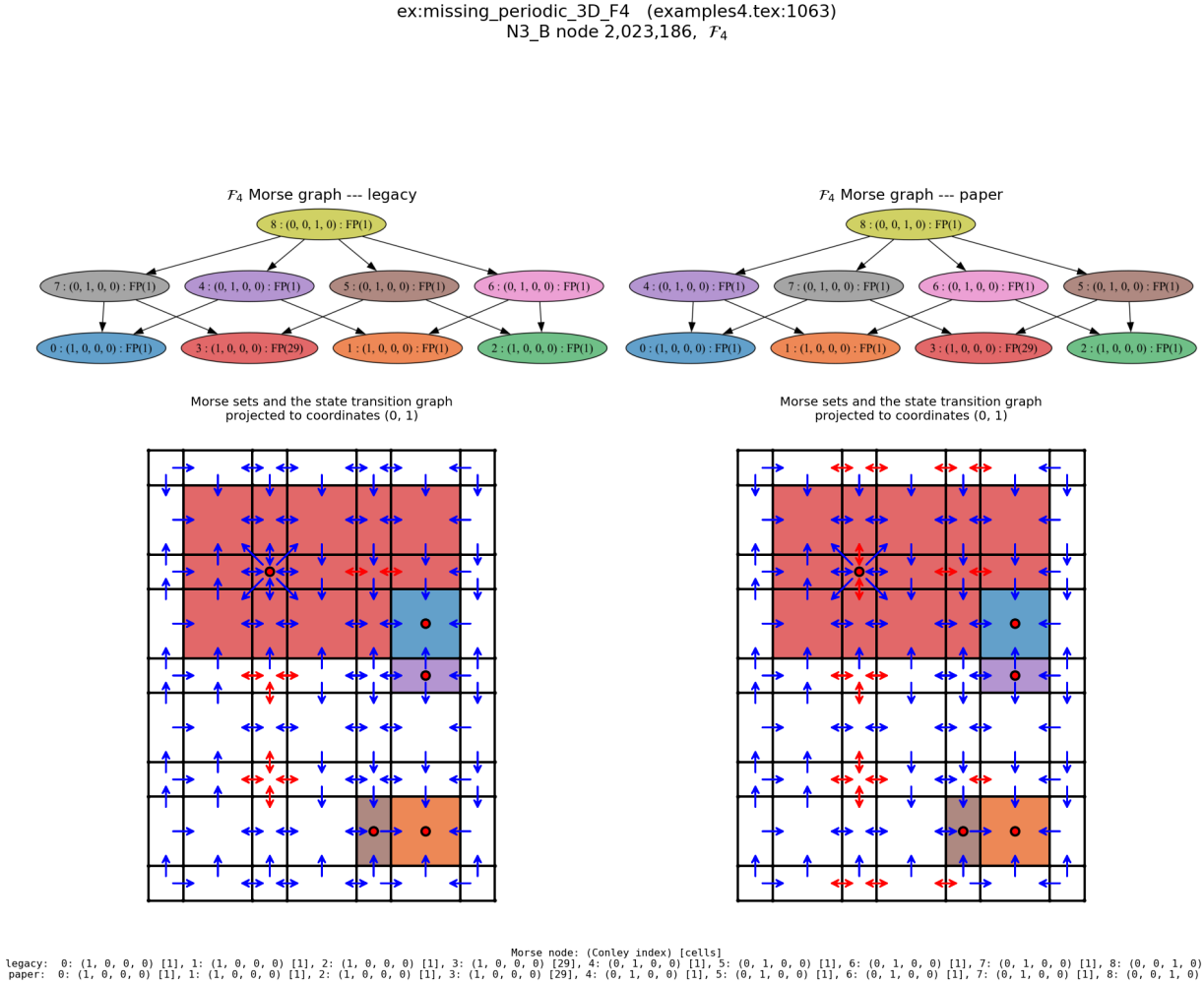


Both specifications agree. `ex:periodicOrbit3_F4` (`examples4.tex:917`), $\mathcal{F}_4$, dimension 3.



Morse node: (Conley index) [cells]
legacy: 0: (1, 1, 0, 0, 0, 0) [12], 1: (1, 0, 0, 0, 0, 0) [1], 2: (1, 0, 0, 0, 0, 0) [1], 3: (0, 0, 1, 0, 0, 0) [1], 4: (0, 1, 0, 0, 0, 0) [1], 5: (0, 1, 0, 0, 0, 0) [1], 6: (0, 1, 0, 0, 0, 0) [1], 7: (0, 0, 1, 0, 0, 0) [1]
paper: 0: (1, 1, 0, 0, 0, 0) [12], 1: (1, 0, 0, 0, 0, 0) [1], 2: (1, 0, 0, 0, 0, 0) [1], 3: (0, 0, 1, 0, 0, 0) [1], 4: (0, 1, 0, 0, 0, 0) [1], 5: (0, 1, 0, 0, 0, 0) [1], 6: (0, 1, 0, 0, 0, 0) [1], 7: (0, 0, 1, 0, 0, 0) [1]

Both specifications agree. `ex:periodicOrbit5` (`examples4.tex:979`), $\mathcal{F}_4$, dimension 5.



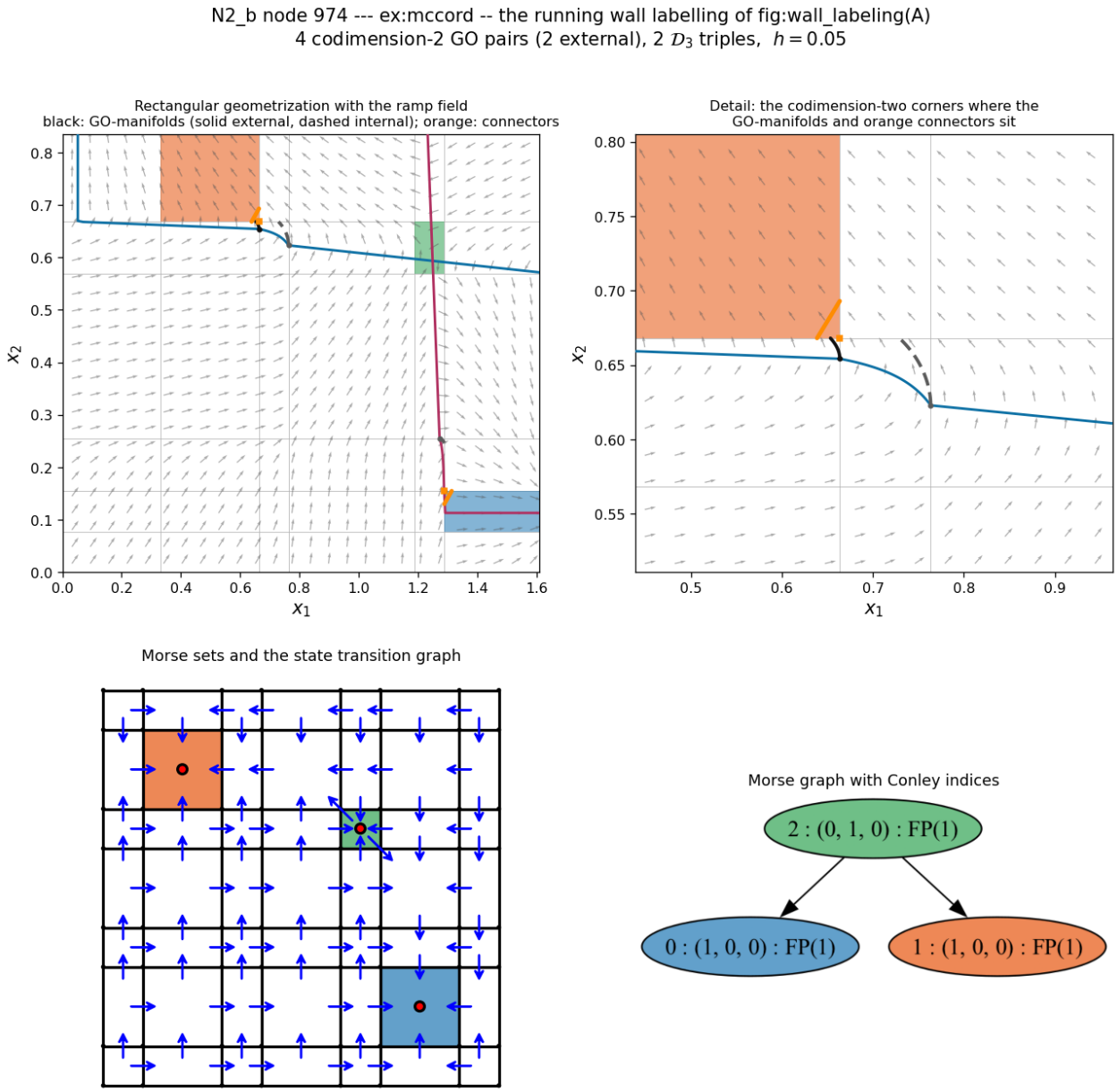
Both specifications agree. `ex:missing_periodic_3D_F4` (`examples4.tex:1063`), $\mathcal{F}_4$, dimension 3.

Two-dimensional geometrizations

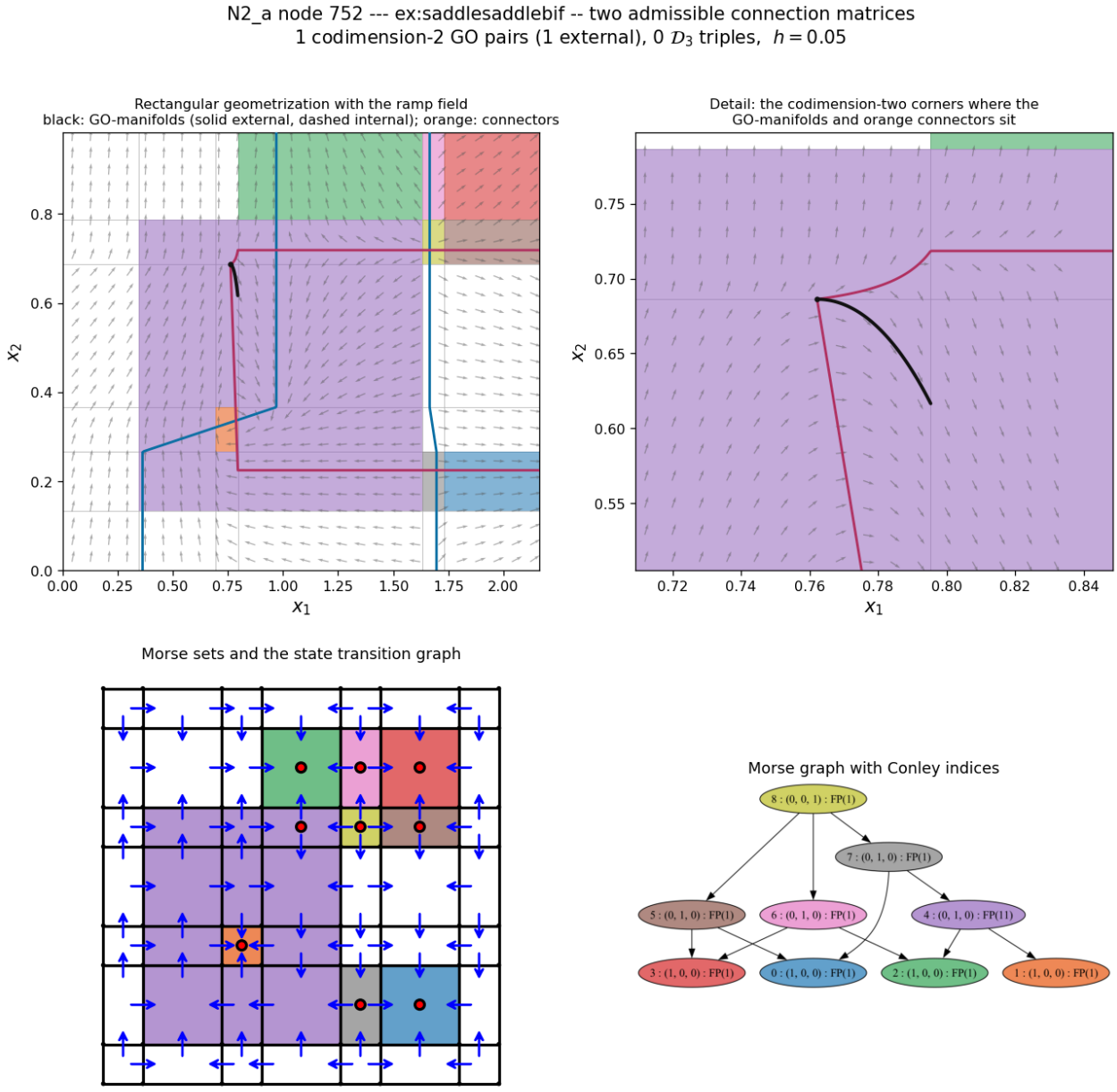
Each DSGRN parameter node below is realised by an explicit ramp system through defn:DSGRN_ramp: a point of the semi-algebraic region is sampled, each switching function is replaced by a type-1 ramp with the same plateaus, and the induced wall labelling is checked to be the one DSGRN computes combinatorially — which is the content of prop:DSGRN_wall_constant.

That gives a genuine vector field on the geometrized complex, so the left panel of each figure carries, in real phase-space coordinates: the rectangular geometrization of defn:rectGeoXb (grey grid), the Morse sets (shaded), the ramp field, the two nullclines (blue $\dot{x}_1 = 0$, red $\dot{x}_2 = 0$), the GO-manifolds of defn:GO-manifold (black solid where the pair is externally pruned, grey dashed where internally pruned), and the orange connectors of defn:orange-manifold for the triples of $\mathcal{D}_3(\Phi)$.

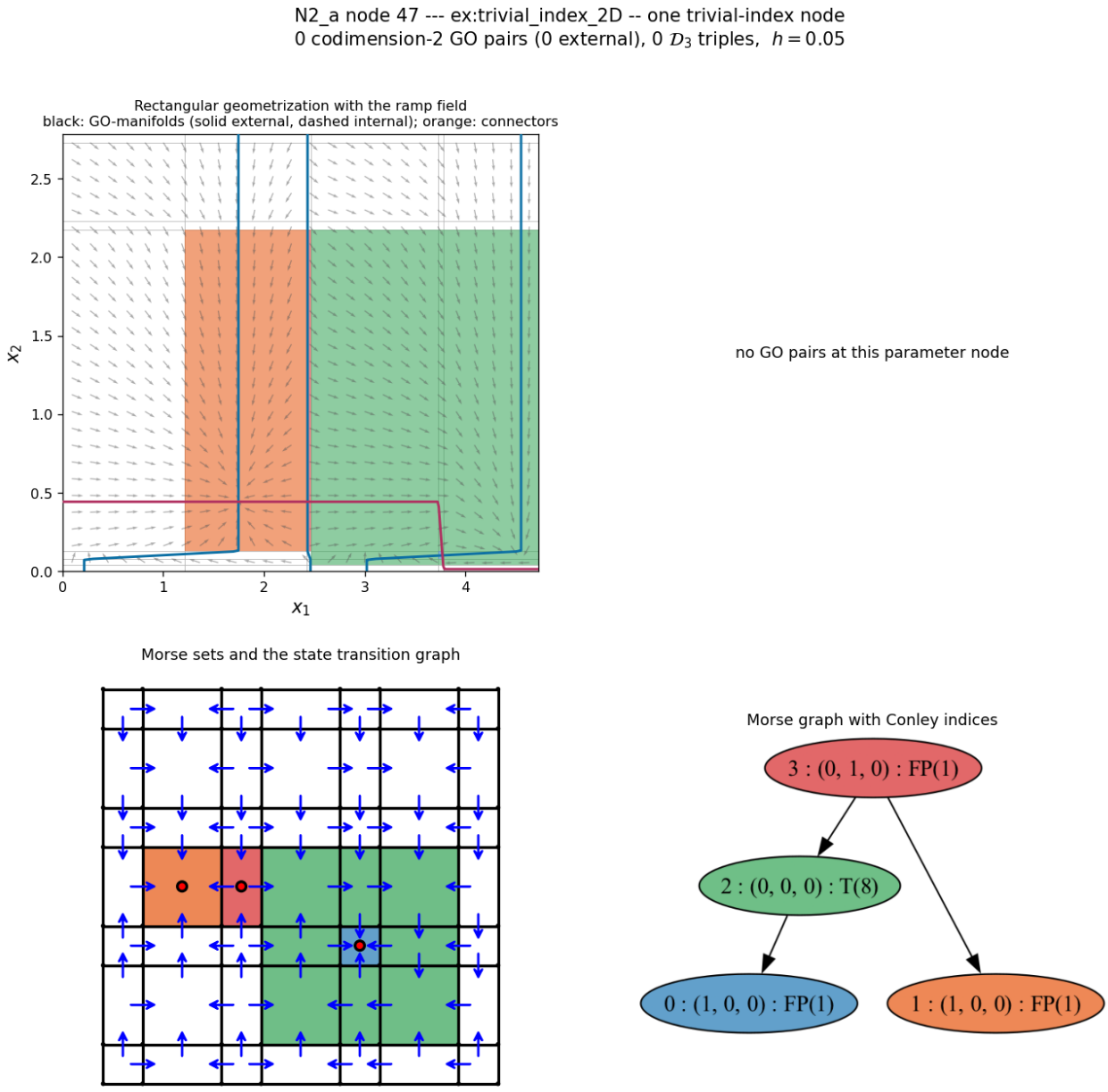
The axes are cropped to the threshold range: the outermost cells of the complex extend to GB_n, which for a sampled DSGRN parameter is far beyond the thresholds.



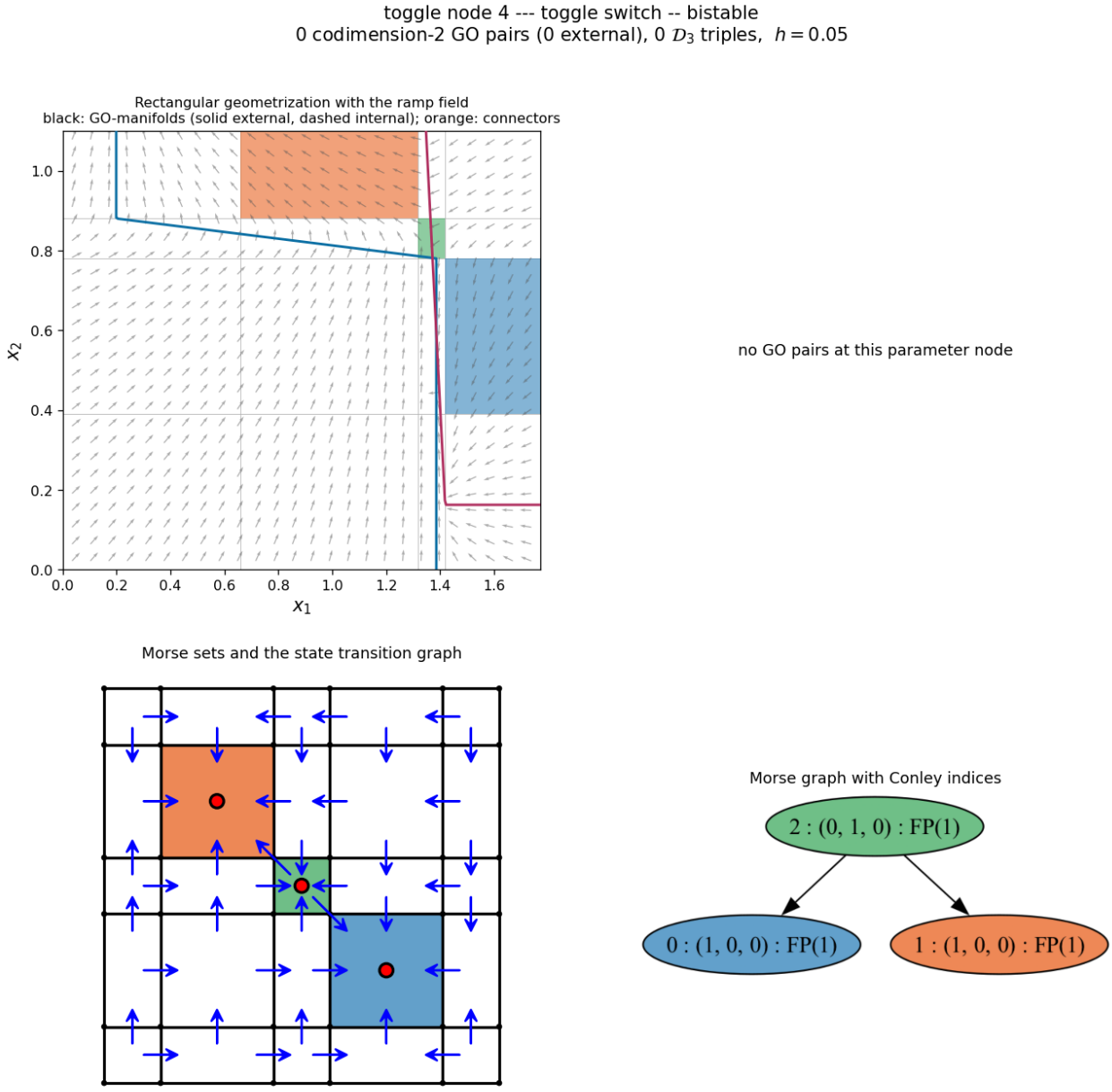
N2_b node 974 — ex:mccord – the running wall labelling of fig:wall_labeling(A). 4 codimension-two GO pairs (2 externally pruned), 2 $\mathcal{D}_3$ triples, 3 Morse nodes.



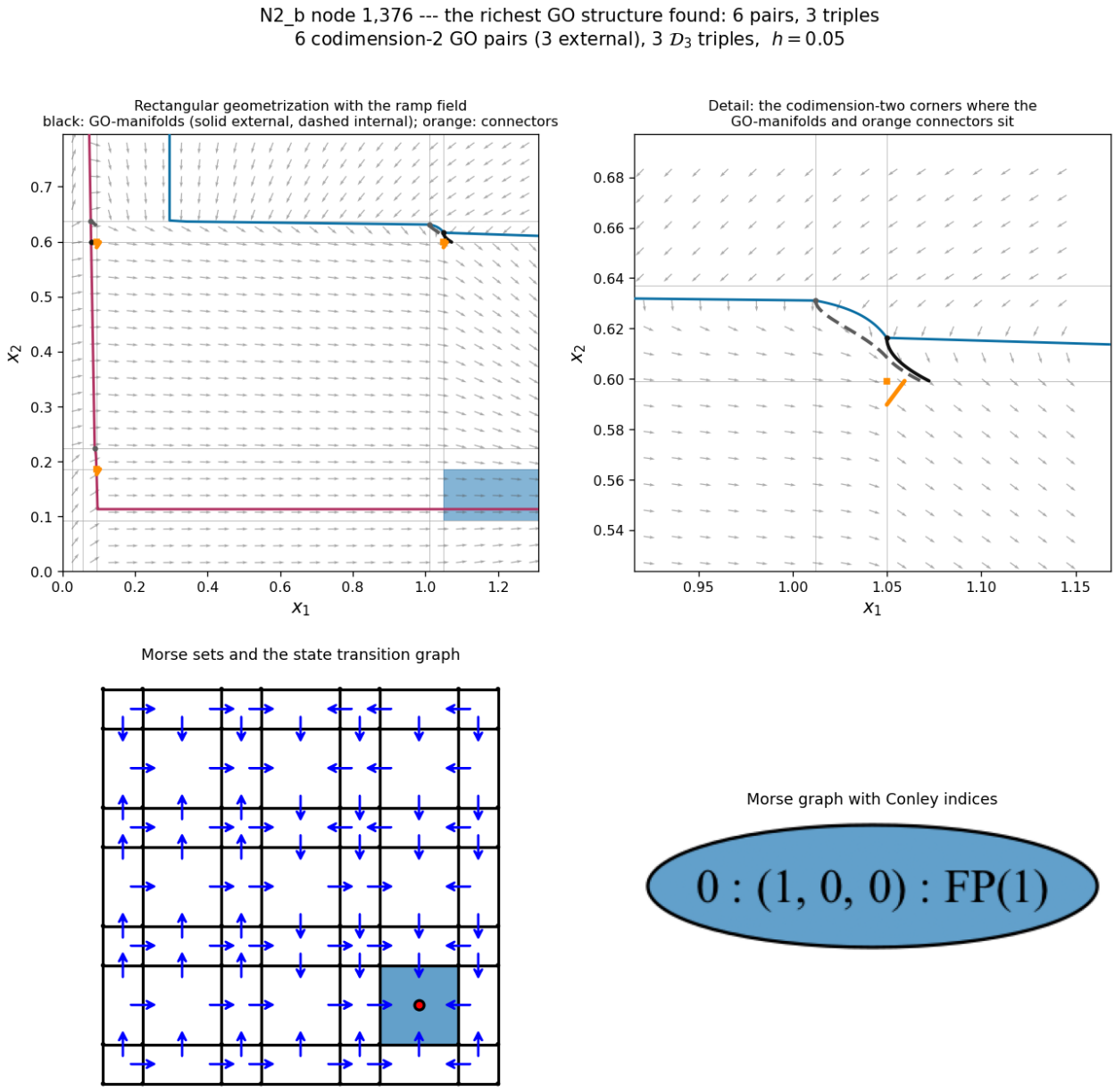
N2_a node 752 — ex:saddlesaddlebif – two admissible connection matrices. 1 codimension-two GO pairs (1 externally pruned), 0 $\mathcal{D}_3$ triples, 9 Morse nodes.



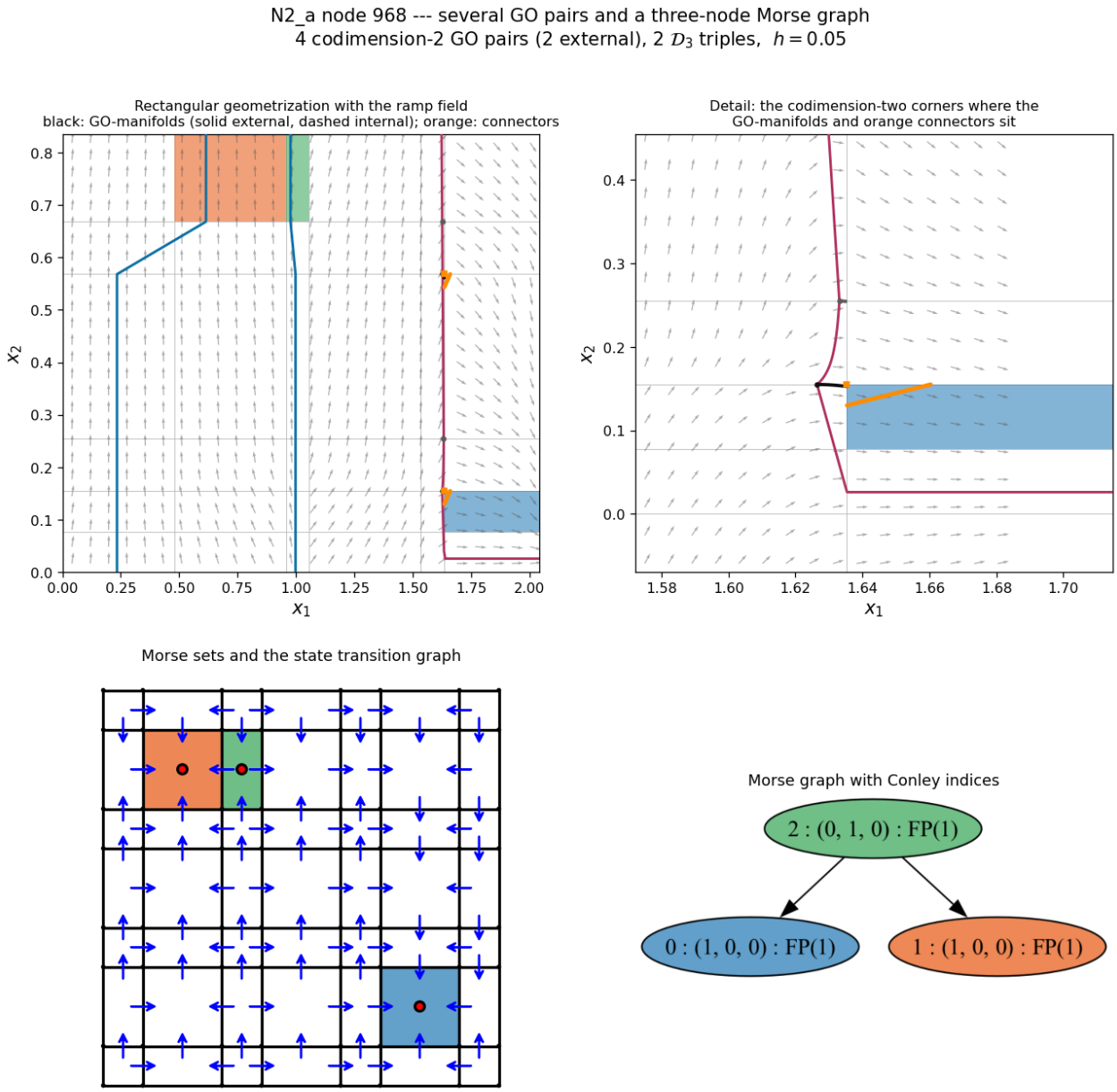
N2_a node 47 — ex:trivial_index_2D – one trivial-index node. 0 codimension-two GO pairs (0 externally pruned), 0 $\mathcal{D}_3$ triples, 4 Morse nodes.



`toggle` node 4 — toggle switch – bistable. 0 codimension-two GO pairs (0 externally pruned), 0 $\mathcal{D}_3$ triples, 3 Morse nodes.

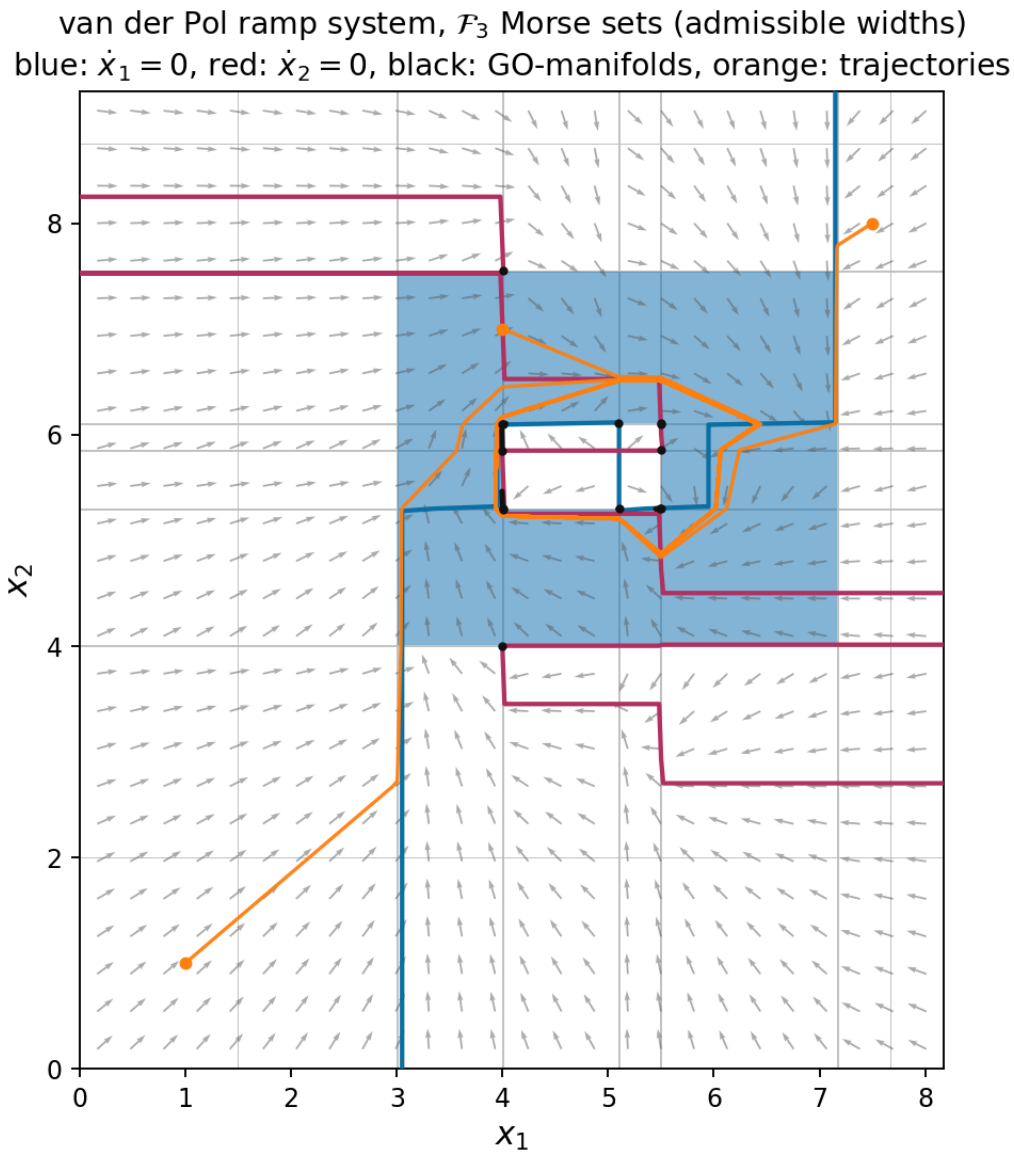


N2_b node 1,376 — the richest GO structure found: 6 pairs, 3 triples. 6 codimension-two GO pairs (3 externally pruned), 3 $\mathcal{D}_3$ triples, 1 Morse nodes.

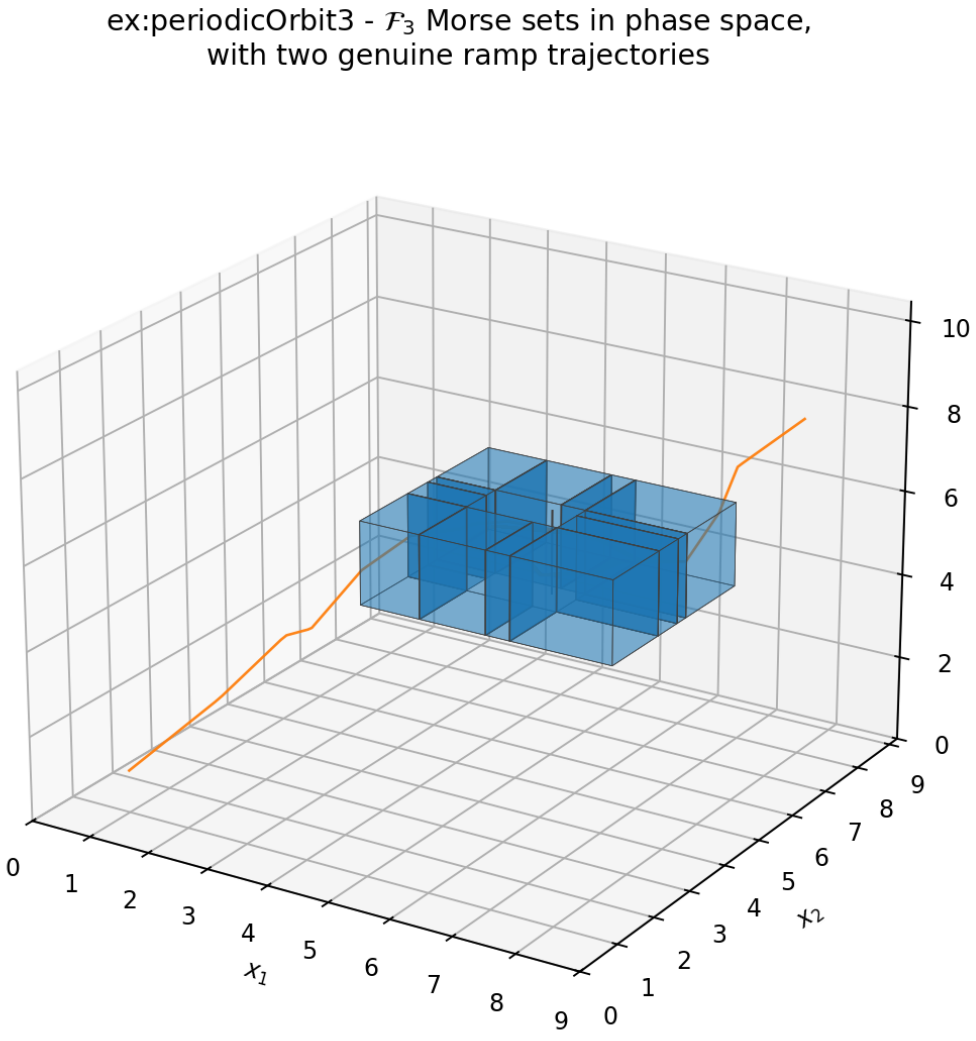


N2_a node 968 — several GO pairs and a three-node Morse graph. 4 codimension-two GO pairs (2 externally pruned), 2 $\mathcal{D}_3$ triples, 3 Morse nodes.

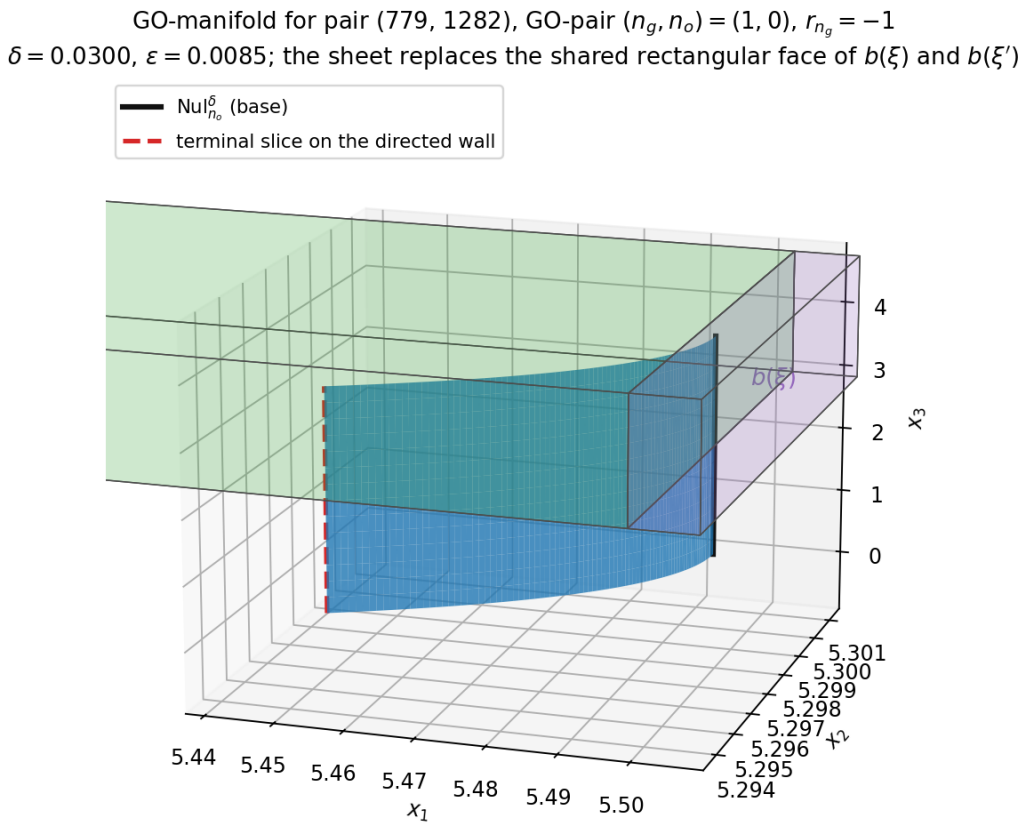
Ramp systems specified directly in the text



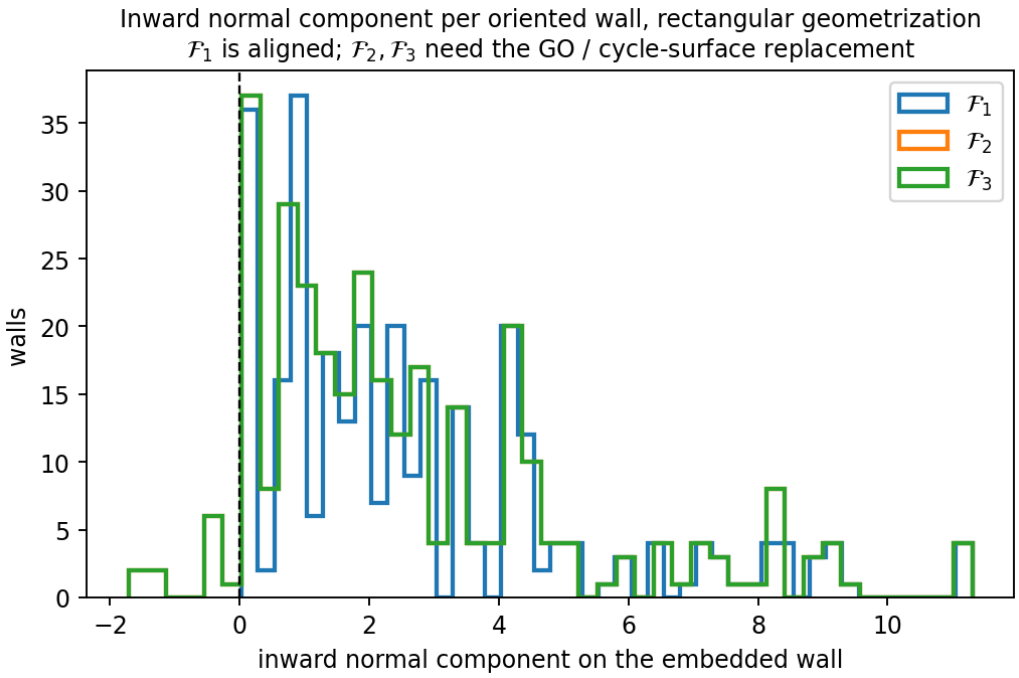
ex:ramp_van_der_pol at an admissible width. Shaded: the $\mathcal{F}_3$ Morse sets, drawn as embedded blowup cells in real ramp coordinates. Blue and red curves: the nullclines $\dot{x}_1 = 0$ and $\dot{x}_2 = 0$. Black: the GO-manifolds of defn:GO-manifold. Orange: genuine trajectories.



ex:periodicOrbit3: the $\mathcal{F}_3$ Morse sets in phase space with two integrated trajectories. The stable periodic orbit is the component with $CH_0 \cong CH_1 \cong \mathbb{Z}_2$.



One GO-manifold, in the context of the two blowup cells whose shared rectangular face it replaces. Black: the δ-nullcline base. Blue sheet: its flow-out under the perturbed field `eq:rampSysPerturbed`. Dashed red: the terminal slice on the directed n_g-wall.



Inward normal component per oriented wall. $\mathcal{F}_1$ lies entirely to the right of zero; the $\mathcal{F}_2$ and $\mathcal{F}_3$ tails to the left are the walls needing the geometric modification.

Reproducing

```
cd code
./.venv/bin/python -m pytest rookfields/tests          # 145 tests
./.venv/bin/python -m rookfields.examples --spec both  # this report's data
./.venv/bin/python -m rookfields.hypotheses
./.venv/bin/python -m rookfields.alignment
./.venv/bin/python -m rookfields.plotting
./.venv/bin/python -m rookfields.report_pdf            # this document
```

Full prose discussion in `code/rookfields/reports/FINDINGS.md`; the actionable items are also filed in `coauthor-todos.md`. No file under `paper/` was modified.